Chapter 1: Numbers and Sequences

Calculus I

August 28, 2024

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Section 1

Numbers and the Least Upper Bound Theorem

Some Notations

In this notes, we will use the following notations:

- $\mathbb{N} = \{1, 2, 3, \ldots\}$: The set of all natural numbers.
- $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$: The set of all integers.
- $\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$: The set of all rational numbers.
- $\mathbb{R} =$ completion of $\mathbb{Q}$: The set of all real numbers.

Some Notations cont'd

Logic symbols:

$$\begin{cases} x \in D, & x \text{ belongs to } D, x \text{ in } D \\ x \notin D, & x \text{ does not belong to } D \\ \forall x, & \text{for all } x, \text{ for any } x \\ \exists x, & \text{there exists } x \\ \exists! x, & \text{there exists a unique } x \\ p \Rightarrow q, & p \text{ implies } q \\ p \Leftrightarrow q, & p \text{ is equivalent to } q, \text{ or } p \text{ holds if and only if } q \text{ holds} \\ a := b, & a \text{ is defined to be } b \end{cases}$$

Some Notations cont'd

Intervals:

For $a, b \in \mathbb{R}$, we write

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}, \quad (a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$$

and

$$[a, \infty) = \{x \in \mathbb{R} \mid x \geq a\}, \quad (-\infty, a] = \{x \in \mathbb{R} \mid x \leq a\}$$

The intervals, $[a, b), (a, b), (a, \infty)$ and $(-\infty, a)$, are defined in a similar way.

Numbers as Infinite Decimals

There are two familiar ways of looking at numbers:

- as infinite decimals
- and as points on the real line.

The integers divide the real line into infinitely many intervals of unit length. If we include the left endpoint of each interval but not the right, we can cover the number line with nonoverlapping intervals such that each number a belongs to exactly one of them, $n \leq a < n + 1$.

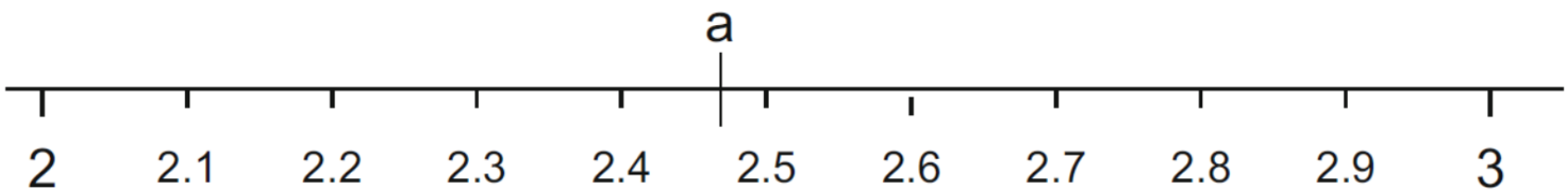
Numbers as Infinite Decimals cont'd

Each interval can be subdivided into ten subintervals of length $\frac{1}{10}$. As before, if we agree to count the left endpoint but not the right as part of each interval, the intervals do not overlap. Our number a belongs to exactly one of these ten subintervals, say to

$$n + \frac{\alpha_1}{10} \leq a < n + \frac{\alpha_1 + 1}{10}$$

This determines the first decimal digit α_1 of a.

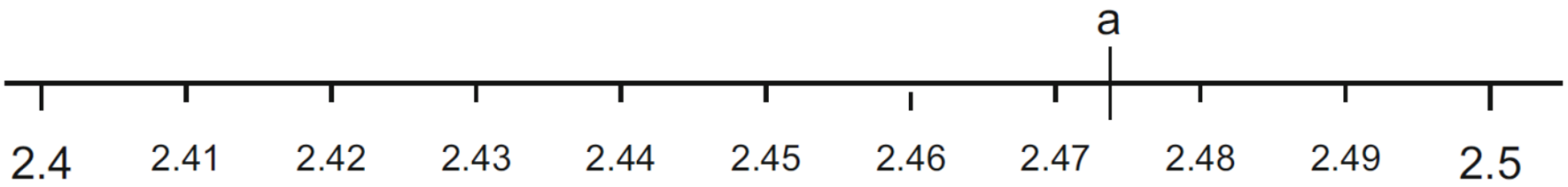
Example.



a is in the interval $[2.4, 2.5)$. So $\alpha_1 = 4$.

Numbers as Infinite Decimals cont'd

The second decimal digit α_2 is determined similarly, by subdividing the interval $[2.4, 2.5)$ into ten equal subintervals, and so on. The following figure illustrates the example $\alpha_2 = 7$



a is in the interval $[2.47, 2.48)$, So $\alpha_2 = 7$.

Thus using the representation of a as a point on the real line and the procedure just described, we can find α_k in $a = n.\alpha_1\alpha_2 \ldots \alpha_k \ldots$ by determining the appropriate interval in the k-th step of this process. Conversely, once we have the decimal representation of a number, we can identify its location on the real line.

Example

Examining the decimal representation $\frac{31}{39} = 0.7948717\ldots$, we see that

$$0.79487 \leq \frac{31}{39} < 0.79488.$$

Repeated Nines in Decimals

The method we described for representing numbers as infinite decimals does not result in decimal fractions that end with infinitely many nines.

Nevertheless, such decimals come up when we do arithmetic with infinite decimals. For instance, take the sum

$$\begin{aligned} 1/3 &= 0.333333333\cdots \\ +2/3 &= 0.666666666\cdots \\ \hline 1 &= 0.999999999\cdots \end{aligned}$$

Similarly, every infinite decimal ending with all nines is equal to a finite decimal, such as

$$0.39529999999\cdots = 0.3953.$$

Decimals and Ordering

The importance of the infinite decimal representation of numbers lies in the ease with which numbers can be compared.

For example, which of the numbers

$$\frac{17}{20}, \quad \frac{31}{39}, \quad \frac{45}{53}, \quad \frac{74}{87}$$

is the largest?

Decimals and Ordering cont'd

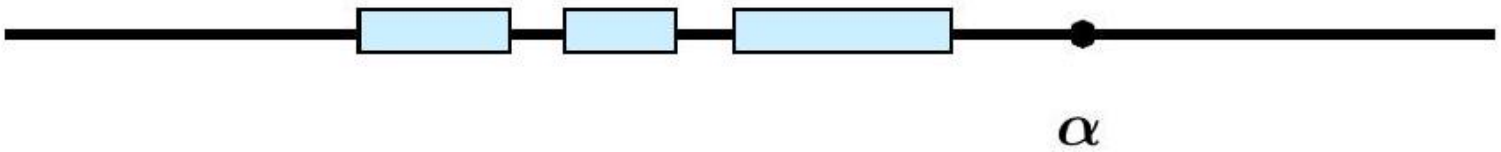
To compare them as fractions, we would have to bring them to a common denominator. If we represent the numbers as decimals,

$$\frac{17}{20} = 0.85000\ldots$$
$$\frac{31}{39} = 0.79487\ldots$$
$$\frac{45}{53} = 0.84905\ldots$$
$$\frac{74}{87} = 0.85057\ldots$$

we can tell which number is larger by examining their integer parts and decimal digits, place by place. Then clearly,

$$\frac{31}{39} < \frac{45}{53} < \frac{17}{20} < \frac{74}{87}.$$

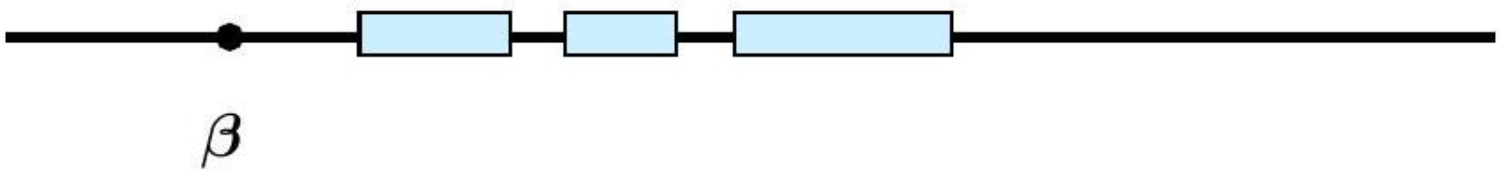
Sets that are bounded above



Let $S \subset \mathbb{R}$.

- S is bounded above if S has an upper bound α.

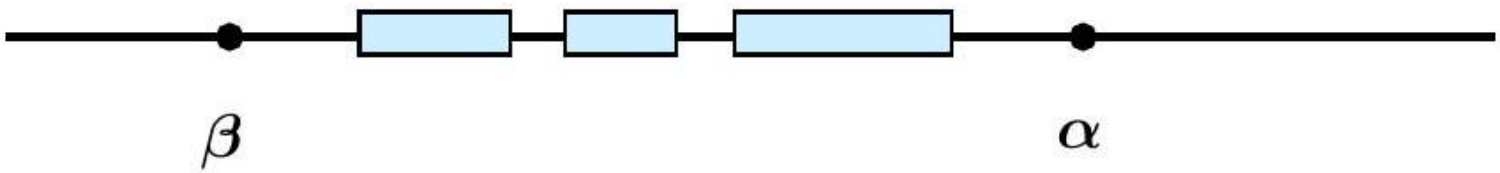
Sets that are bounded below



Let $S \subset \mathbb{R}$.

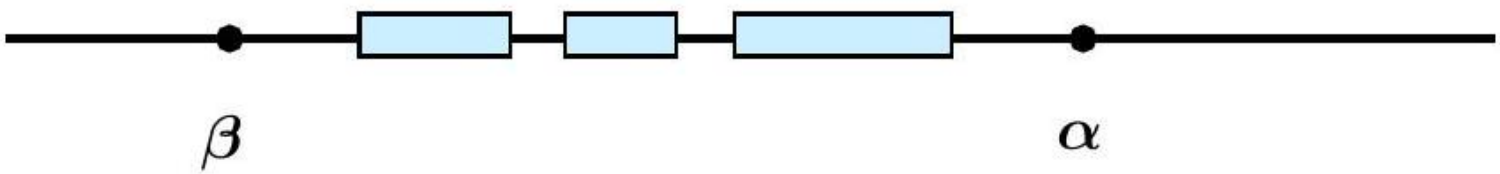
- S is bounded below if S has an lower bound β.

Bounded sets



Let $S \subset \mathbb{R}$.

- S is bounded if S is bound above and bounded below.

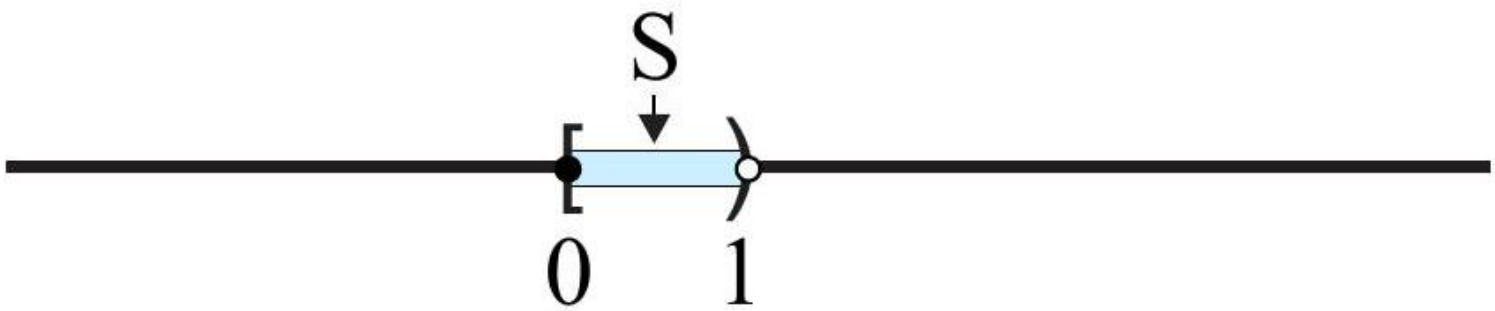


Definition 1.1.1

Let $S \subset \mathbb{R}$.

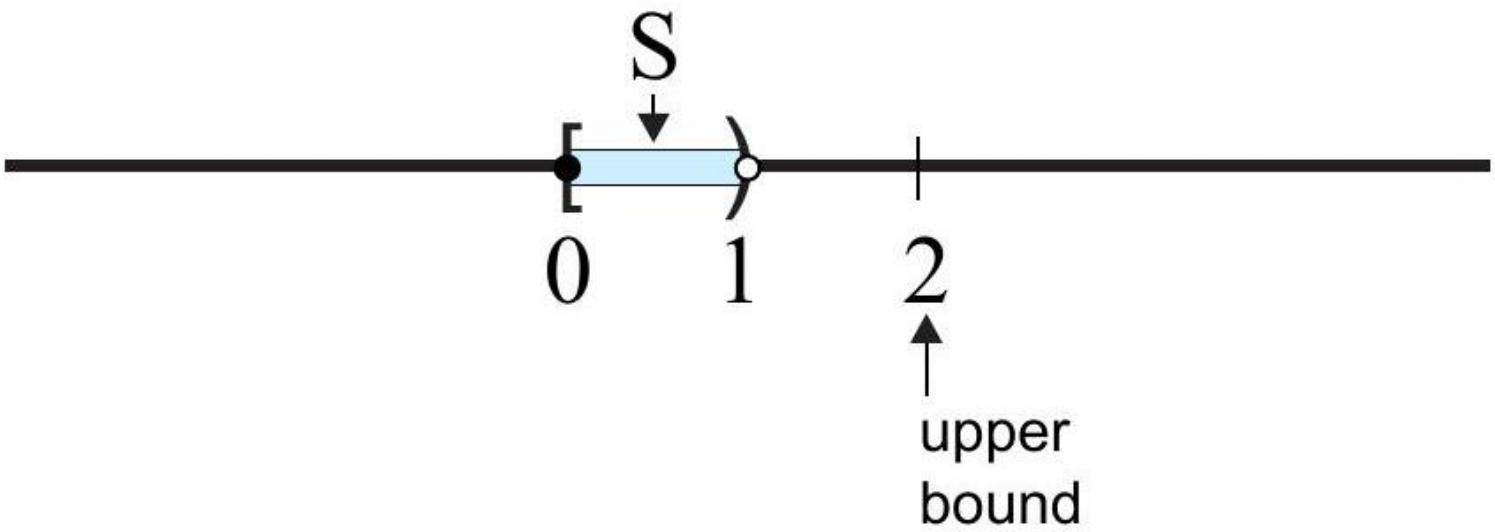
1. S is bounded above if S has an upper bound α.
2. S is bounded below if S has an lower bound β.
3. S is bounded if S is bound above and bounded below.

Example



Let $S = [0, 1)$.

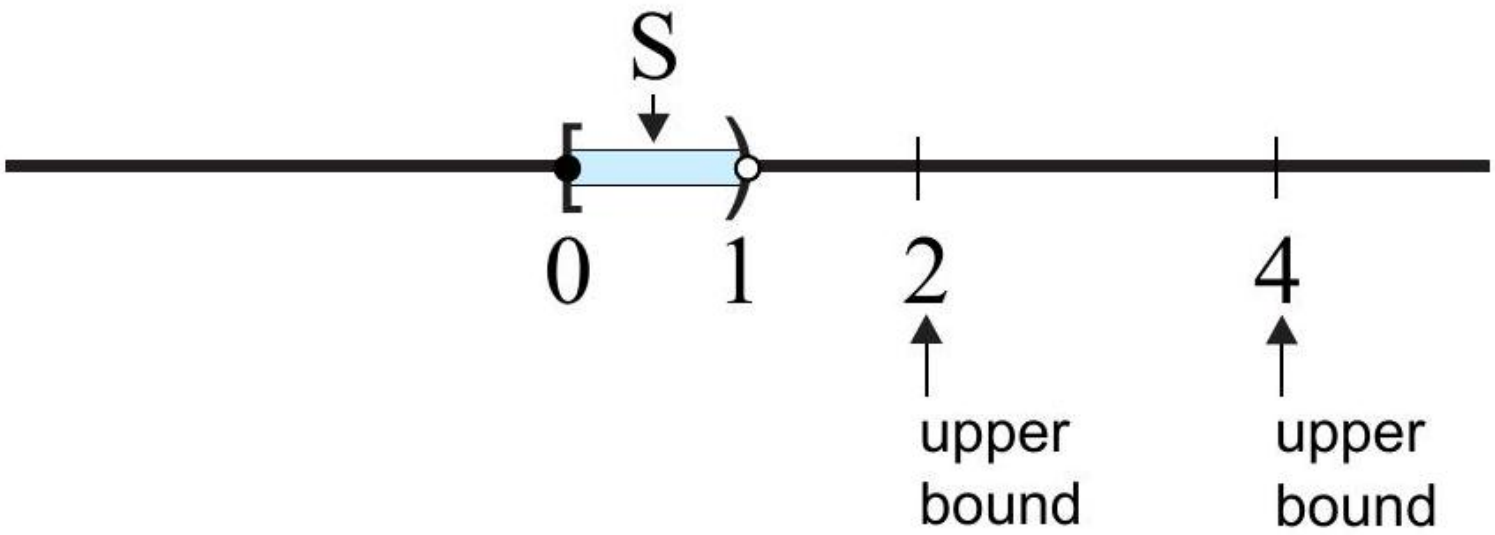
Example cont'd



Let $S = [0, 1)$.

- S is bounded above by 2.

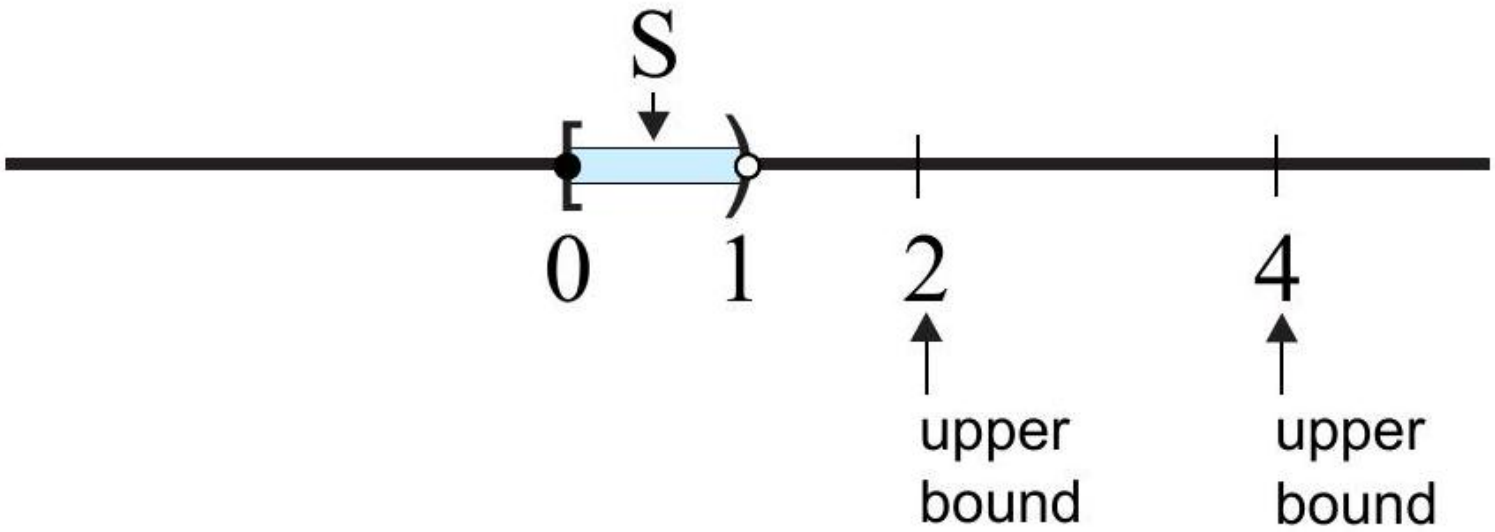
Example cont'd



Let $S = [0, 1)$.

- S is bounded above by 2.
- S is also bounded above by 4.

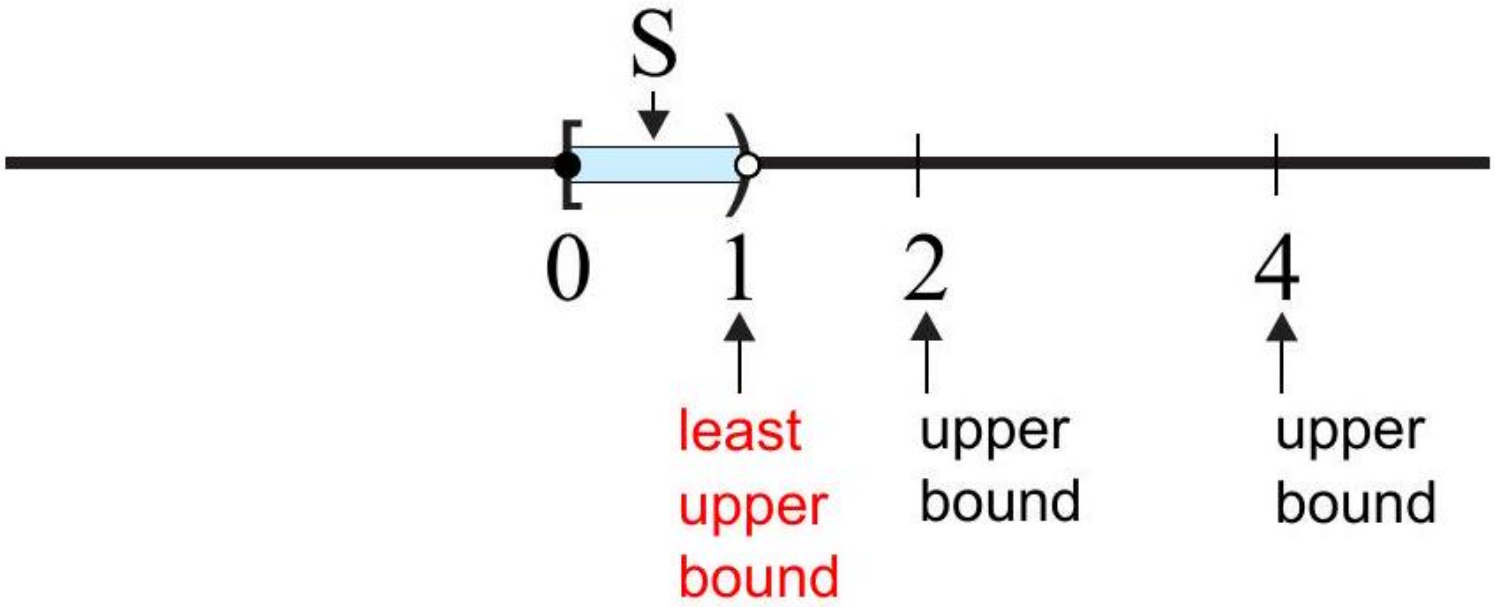
Example cont'd



Let $S = [0, 1)$.

- S is bounded above by 2.
- S is also bounded above by 4.
- S has infinitely many upper bounds.

Example cont'd



Let $S = [0, 1)$.

- S is bounded above by 2.
- S is also bounded above by 4.
- S has infinitely many upper bounds.
- 1 is a special upper bound for S.
- 1 is the smallest or least upper bound for S.

Least Upper Bound

Definition 1.1.2 (Least Upper Bound)

We say that $\alpha \in \mathbb{R}$ is the least upper bound for a set $S \subset \mathbb{R}$ if

1. α is an upper bound for S, and
2. if γ is an upper bound for S, then $\alpha \leq \gamma$.

If a set S has a least upper bound, then we denote it by $\mathrm{lub}(S)$.

The least upper bound of S is often called the supremum of S, denoted by $\sup(S)$.

The Least Upper Bound Theorem

Problem: Does every set S that is bounded above have a LUB?

Theorem 1.1.3 (Least Upper Bound Theorem or LUBT)

A nonempty subset $S \subset \mathbb{R}$ that is bounded above always has a least upper bound.

Proof of the LUBT

- We prove the theorem when S is an infinite set of numbers from $[0, 1)$ (the case where S is a finite set is trivial). The proof of the general case is similar.

- Examine the first decimal digits of the numbers in S and keep only those with the largest first digit. We call the remaining numbers eligible after the first step.

- Examine the second digits of the numbers that were eligible after the first step and keep only those with the largest second digit. Those are the numbers that are eligible after the second step. Define the number s by setting its j-th digit equal to the j-th digit of any number that remains eligible after j steps. By construction, s is greater than or equal to every number in S, i.e., s is an upper bound of S.

Proof of the LUBT cont'd

- Next we show that every number that is smaller than s is not an upper bound of S, i.e., s is the smallest, or least, upper bound of S.
- Let $m = 0.m_1m_2m_3 \ldots m_n \ldots$ be any number smaller than $s = 0.s_1s_2s_3 \ldots s_n \ldots$
- Denote by j the first position in which the digits of s and m differ. That means that for $n < j, s_n = m_n$. Since m is smaller than $s, m_j < s_j$.
- At the j-th step in our construction of s there was at least one number x in S that agreed with s up through the j-th decimal digit.
- By comparing decimal digits, we see that m is less than x. So m is not an upper bound for S.
- Since no number less than s is an upper bound for S, s is the least upper bound of S. The proof is complete.

Greatest Lower Bound

Definition 1.1.4 (Greatest Lower Bound)

We say that $\beta \in \mathbb{R}$ is the greatest lower bound for a set $S \subset \mathbb{R}$ if

1. β is a lower bound for S, and
2. if γ is a lower bound for S, then $\beta \geq \gamma$.

If a set S has a greatest bound, then we denote it by $\mathrm{glb}(S)$.

The greatest lower bound of S is often called the infimum of S, denoted by $\inf(S)$.

Theorem 1.1.5 (Greatest Lower Bound Property)

Every set of numbers that is bounded below has a greatest lower bound.

Section 2

Sequences and their Limits

What is a Sequence?

Informally, a sequence is an ordered list of real numbers.

For example a phone number 519-555-1234 is a finite sequence of 1-digit numbers

$$5 - 1 - 9 - 5 - 5 - 5 - 1 - 2 - 3 - 4.$$

Note: Order is important!

During our discussions, sequences will be infinite ordered lists of real numbers of the form $\{a_1, a_2, a_3, \ldots, a_n, \ldots\}$.

What is a Sequence?

More formally:

Definition 1.2.1 (Sequence)

A sequence is a function $f : \mathbb{N} \to \mathbb{R}$.
For $n \in \mathbb{N}$, we call $f(n)$ the n-th term of the sequence and n is called the index.

Notation: We often write a_n or b_n or x_n instead of $f(n)$ to denote the n-th term.

Specifying a Sequence

A sequence can be specified by:

1. Providing the explicit function:
   - $f(n) = \frac{1}{n}$
   - $a_n = \frac{1}{n}$
   - $\left\{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots\right\}$
   - $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ or simply $\left\{\frac{1}{n}\right\}$

Note: Some references use round brackets instead of curly brackets to denote sequences. As such, you may encounter a sequence written as $(a_1, a_2, a_3, \cdots, a_n, \cdots)$ or $(a_n)_{n=1}^{\infty}$ or (a_n).

Specifying a Sequence cont'd

2. Recursion:

Future terms in a sequence are defined from existing terms.

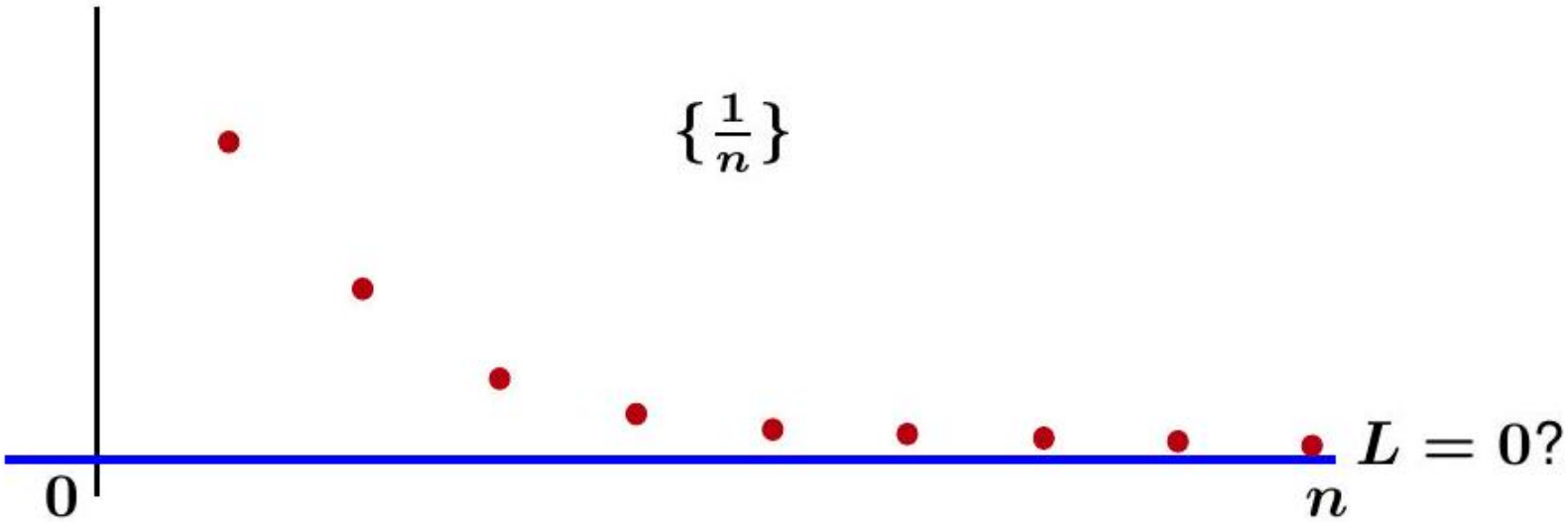
Example: Let $a_1 = 1$ and $a_{n+1} = \frac{1}{1+a_n}$.

What is a_5?

- $a_1 = 1$
- $a_2 = \frac{1}{1+a_1} = \frac{1}{1+1} = \frac{1}{2}$
- $a_3 = \frac{1}{1+a_2} = \frac{1}{1+\frac{1}{2}} = \frac{2}{3}$
- $a_4 = \frac{1}{1+a_3} = \frac{1}{1+\frac{2}{3}} = \frac{3}{5}$
- $a_5 = \frac{1}{1+a_4} = \frac{1}{1+\frac{3}{5}} = \frac{5}{8}$

Exercise: Can you find an explicit formula for a_n?

What is the Limit of a Sequence?

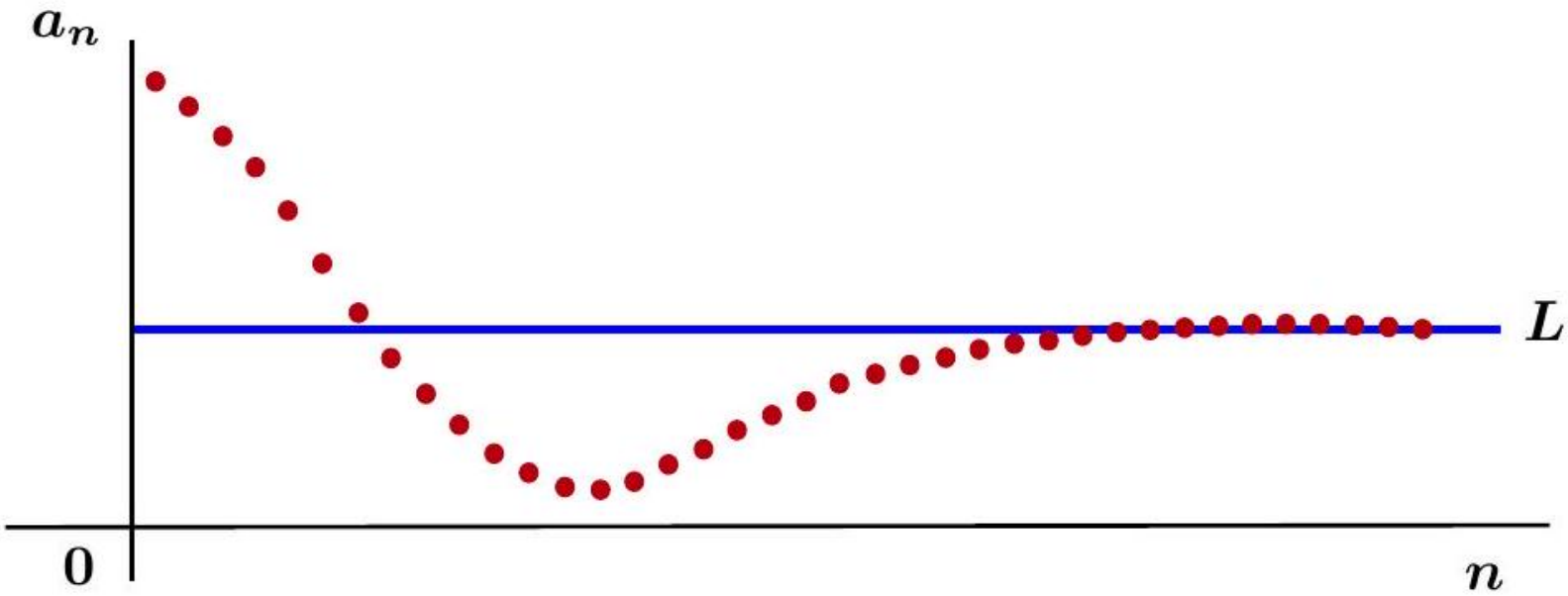


Consider $\left\{\frac{1}{n}\right\}$.

As n gets larger and larger, the terms get closer and closer to 0.

We want to call 0 the limit of the sequence $\left\{\frac{1}{n}\right\}$ as n goes to ∞.

Heuristic Definition of a Limit of a Sequence

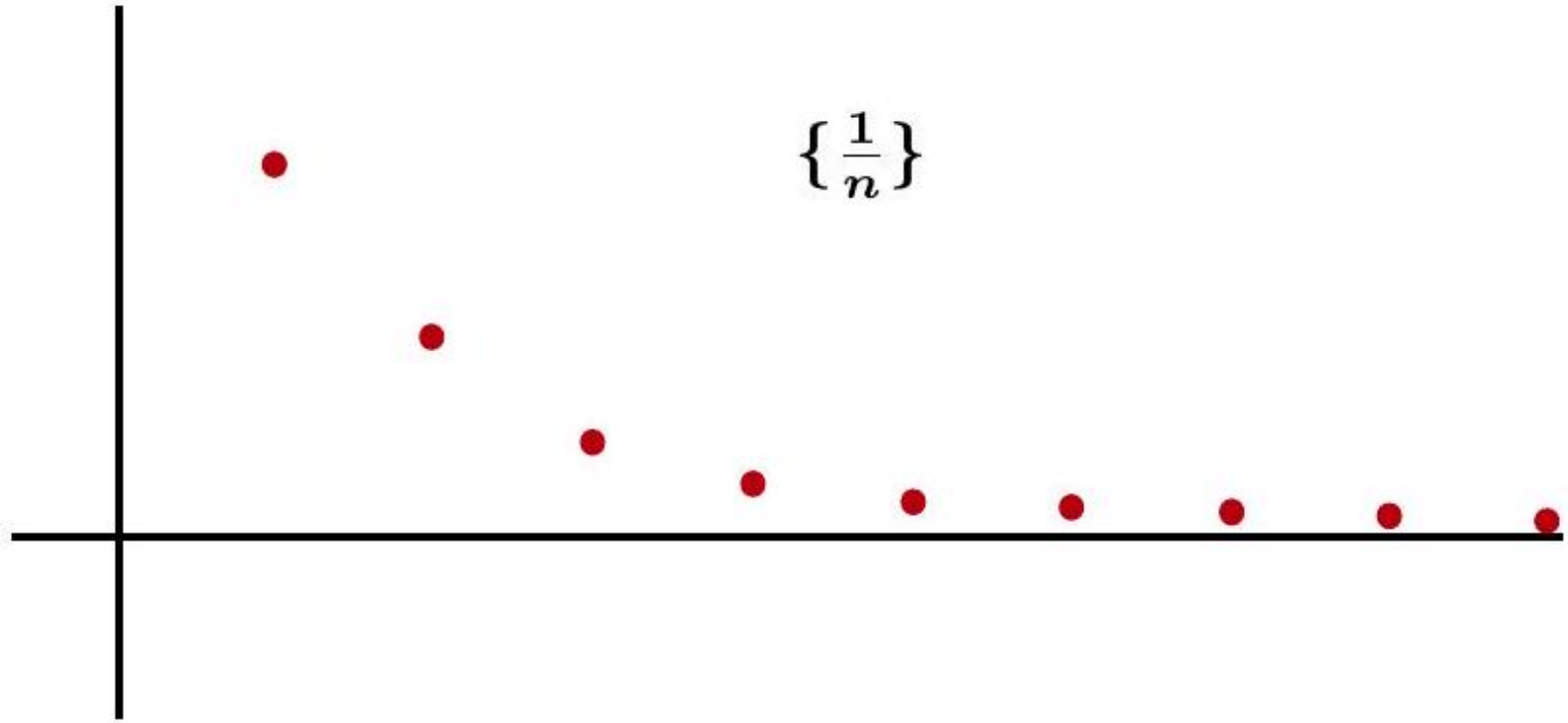


Heuristic Definition:

We say that L is the limit of the sequence $\{a_n\}$ as n goes to ∞ if as n gets larger and larger the terms of $\{a_n\}$ get closer and closer to L.

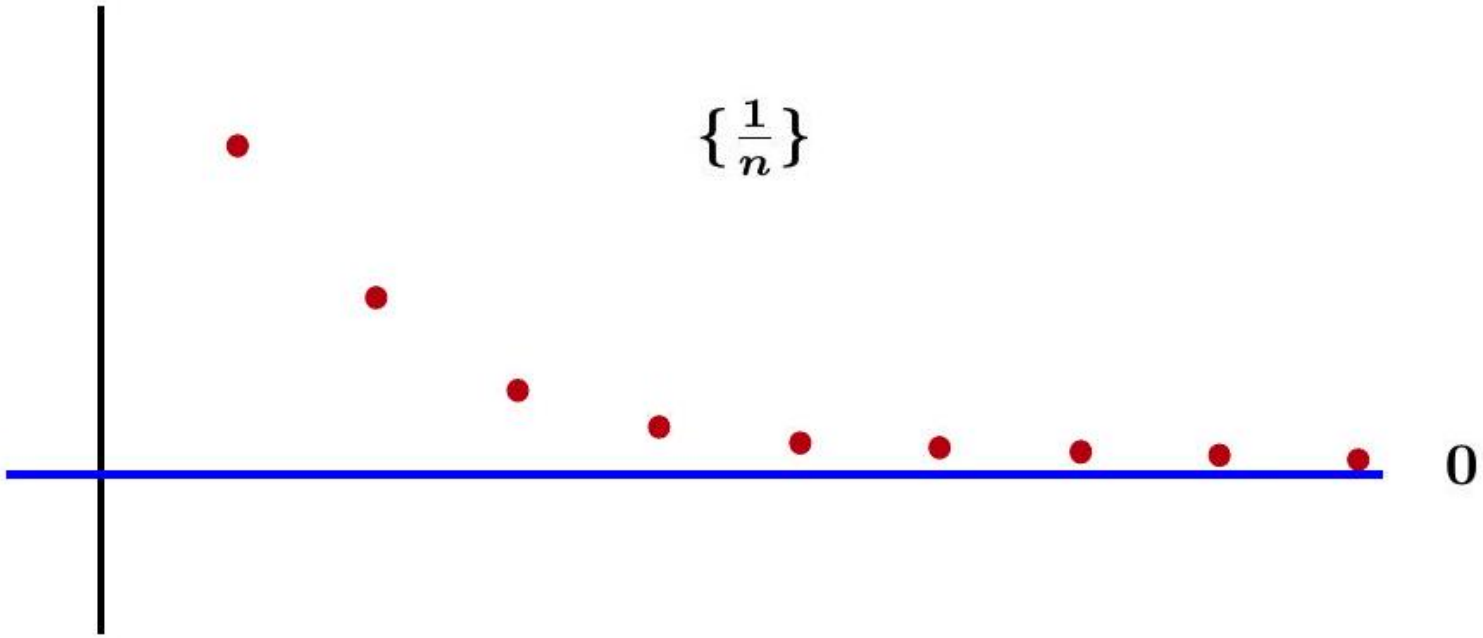
Question: What's wrong with this definition?

Heuristic Definition of a Limit of a Sequence



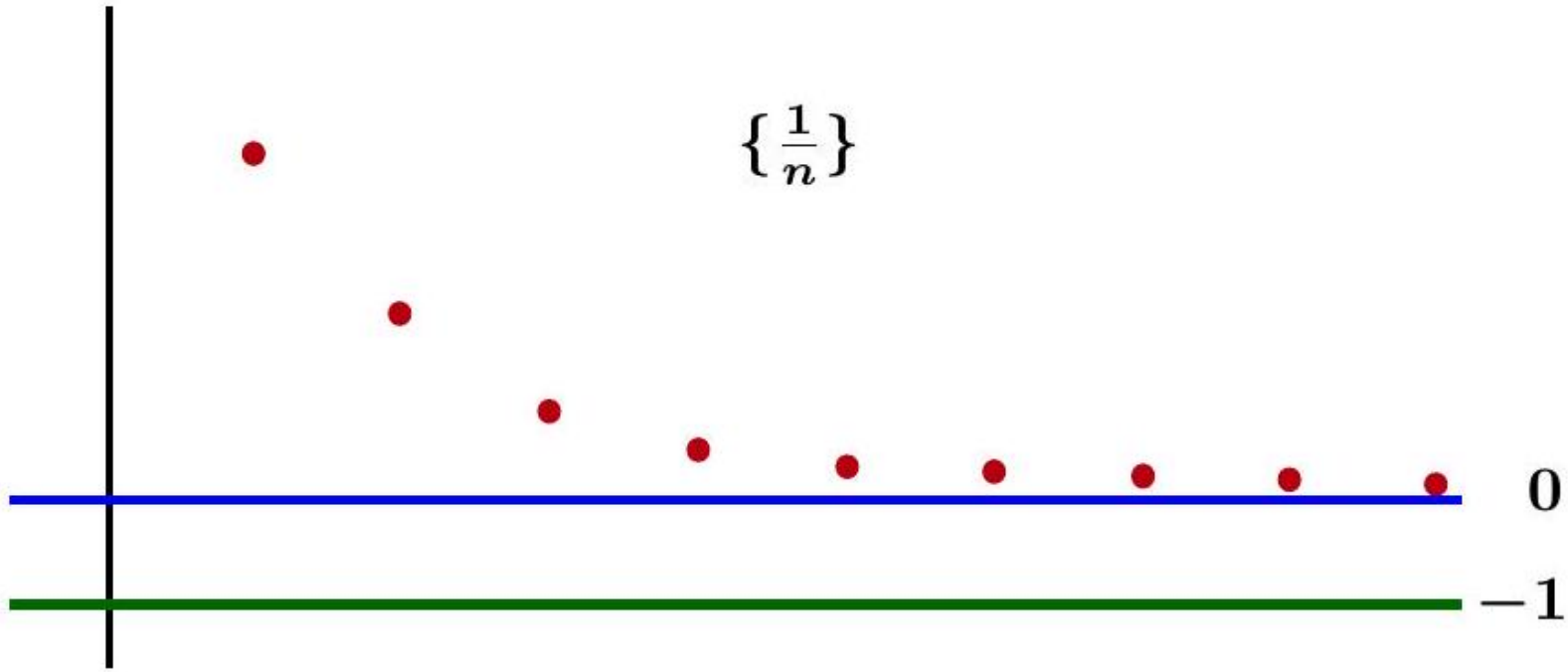
Again, consider $\left\{\frac{1}{n}\right\}$.

Heuristic Definition of a Limit of a Sequence



Again, consider $\left\{\frac{1}{n}\right\}$. As n gets larger and larger, the terms get closer and closer to 0.

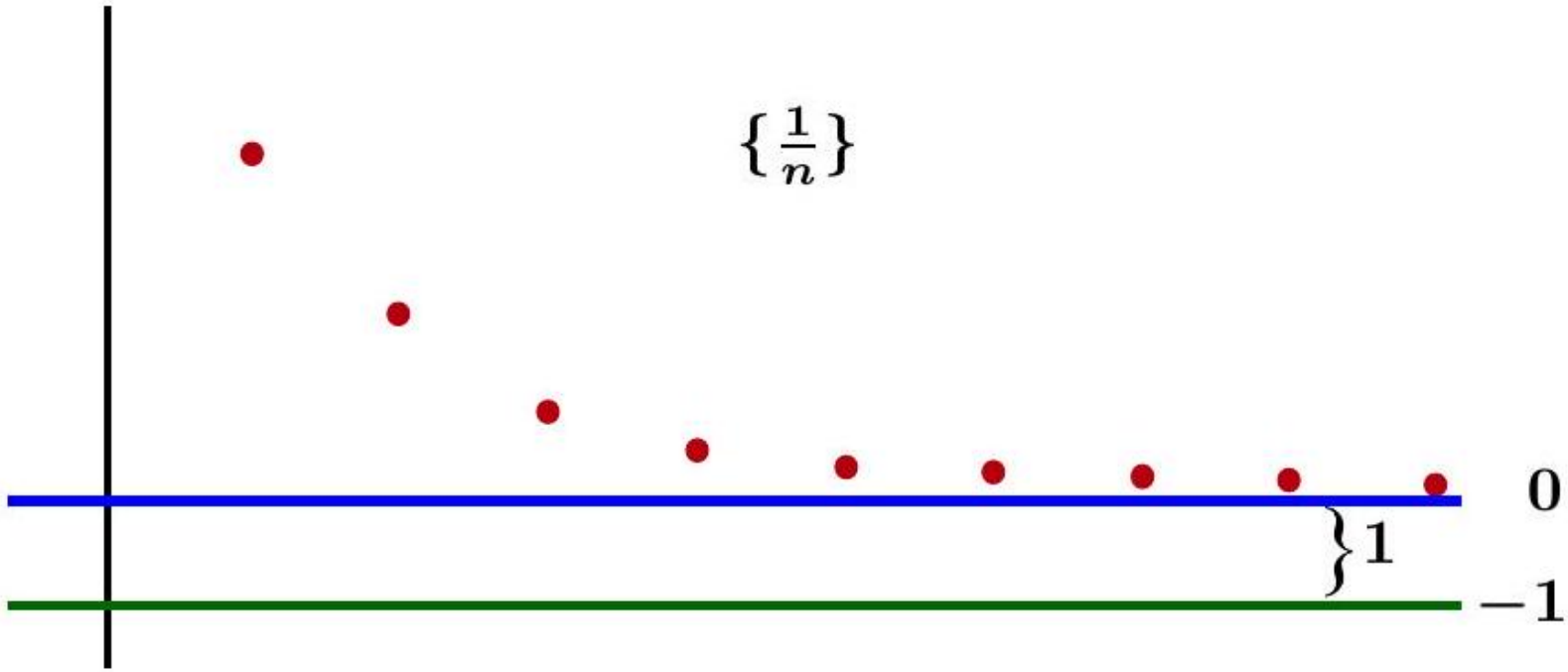
Heuristic Definition of a Limit of a Sequence



Again, consider $\left\{\frac{1}{n}\right\}$. As n gets larger and larger, the terms get closer and closer to 0.

But these terms also get closer and closer to -1.

Heuristic Definition of a Limit of a Sequence



Question: What is special about 0 that makes us choose it as the limit of $\{\frac{1}{n}\}$ instead of -1?

Answer: The terms of $\{\frac{1}{n}\}$ approximate 0 as closely as we would like when n is large enough, but these terms are never within 1 unit of -1 .

New Heuristic Definition:

We say that L is the limit of the sequence $\{a_n\}$ as n goes to infinity if no matter what positive tolerance $\epsilon > 0$ we are given, we can find a cutoff $N \in \mathbb{N}$ such that the terms a_n approximate L with an error less than ϵ provided that $n \geq N$.

Formal Definition

Definition 1.2.2 (Limit of a Sequence)

We say that L is the limit of the sequence $\{a_n\}$ as n goes to infinity if for every $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that if $n \geq N$, then

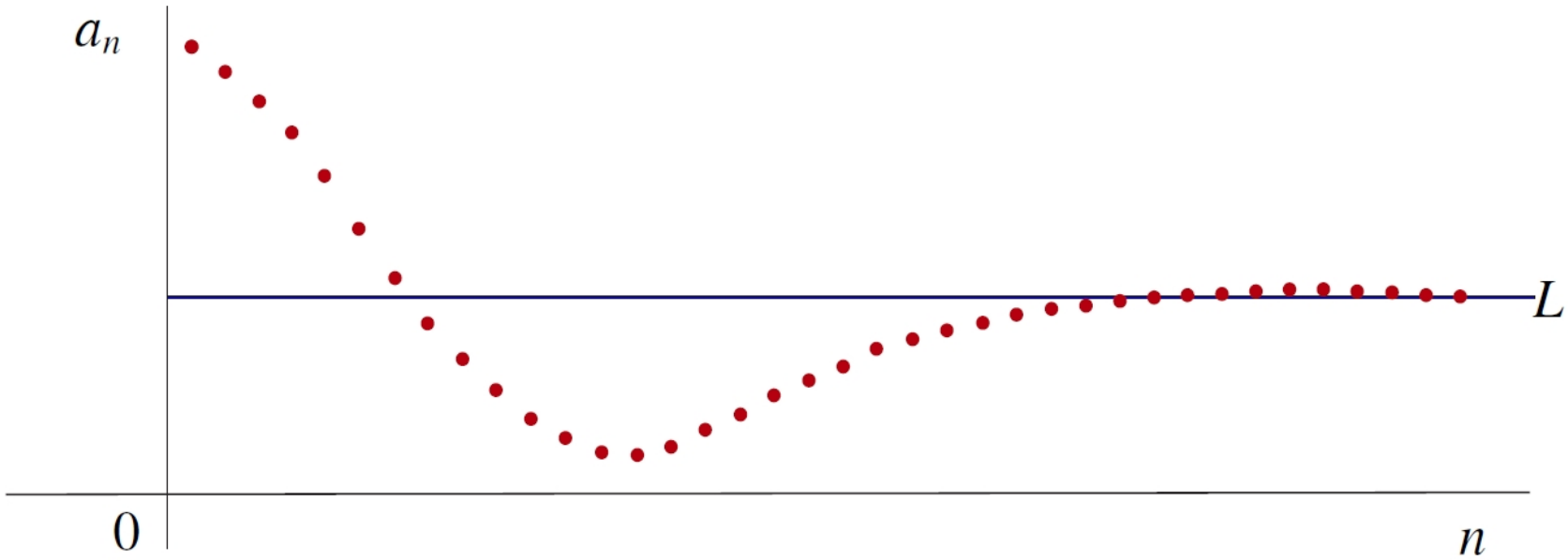
$$|a_n - L| < \epsilon$$

In this case, we say that $\{a_n\}$ converges to L and we write

$$\lim_{n \to \infty} a_n = L.$$

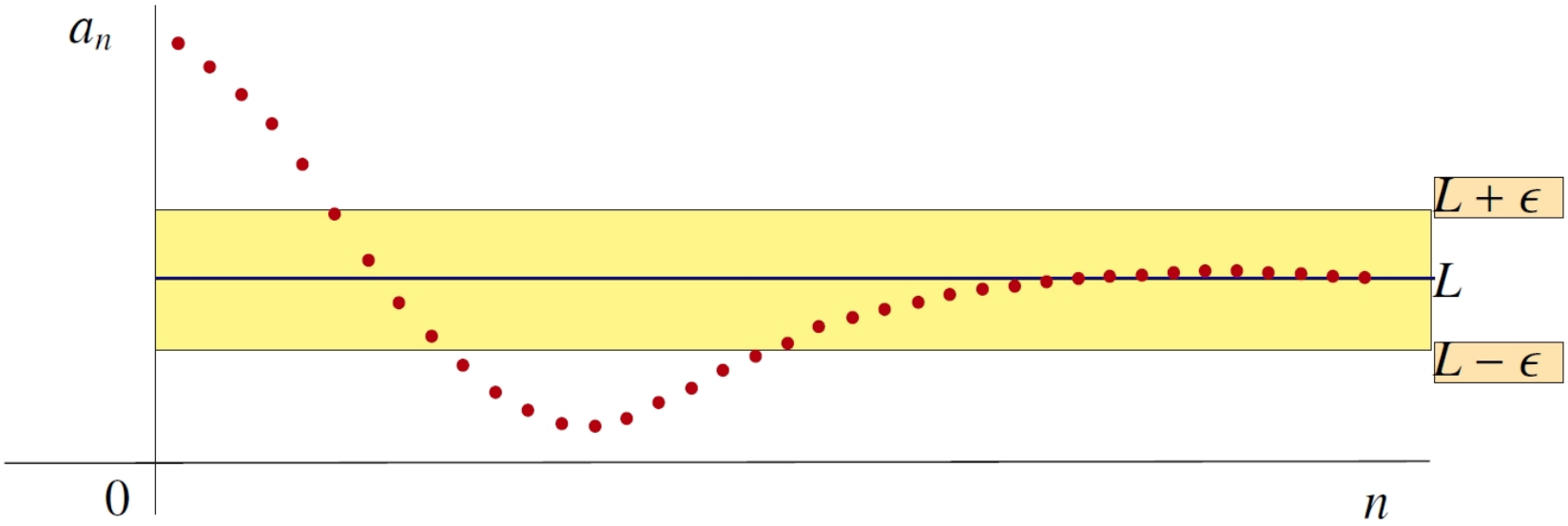
If no such L exists we say that $\{a_n\}$ diverges, or simply the limit of $\{a_n\}$ does not exist.

The Limit of a Sequence



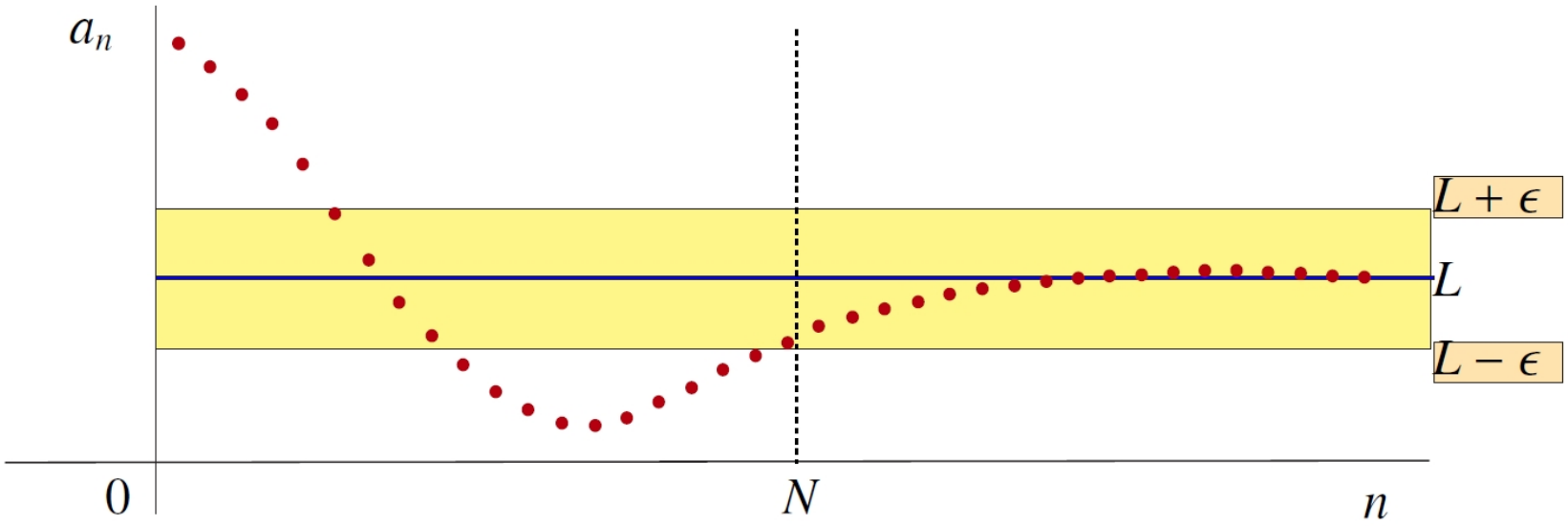
1. Identify L.

The Limit of a Sequence



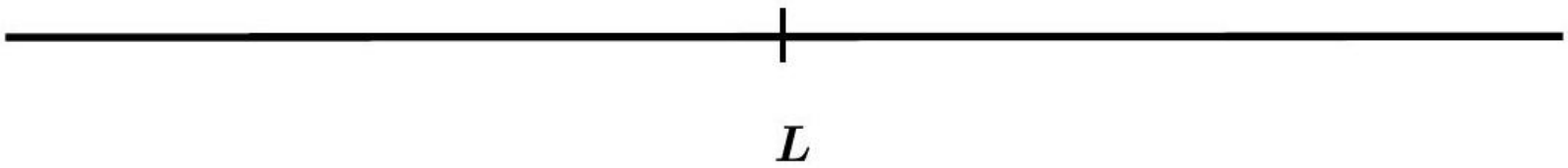
1. Identify L.
2. Specify an arbitrary error $\epsilon > 0$.

The Limit of a Sequence



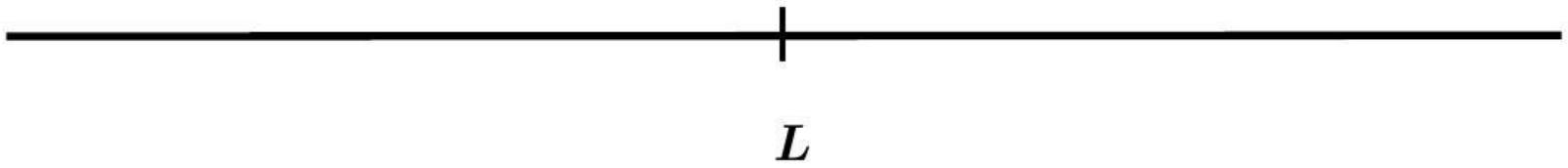
1. Identify L.
2. Specify an arbitrary error $\epsilon > 0$.
3. Find the cutoff N.

The Limit of a Sequence



It is useful to look at how this works on the real line.

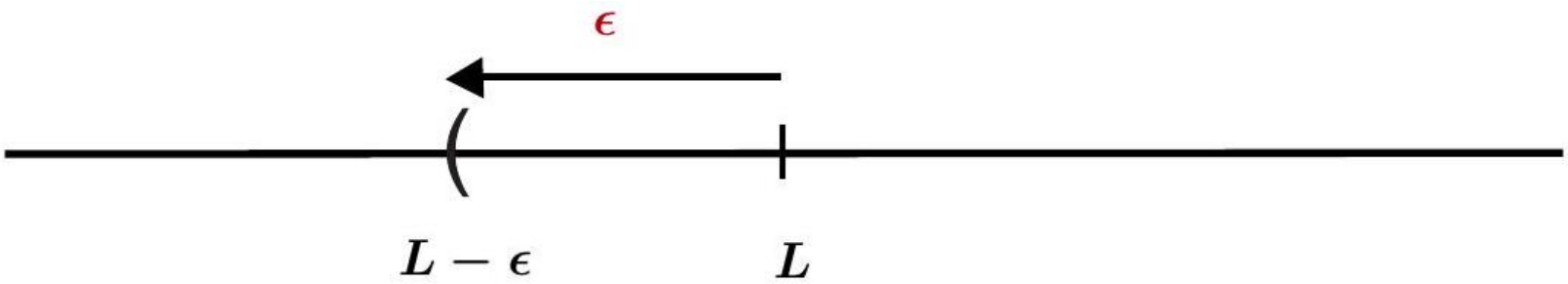
The Limit of a Sequence



It is useful to look at how this works on the real line.

Assume $\lim_{n\to\infty} \boldsymbol{a}_n = \boldsymbol{L}$ and let $\boldsymbol{\epsilon} > 0$.

The Limit of a Sequence

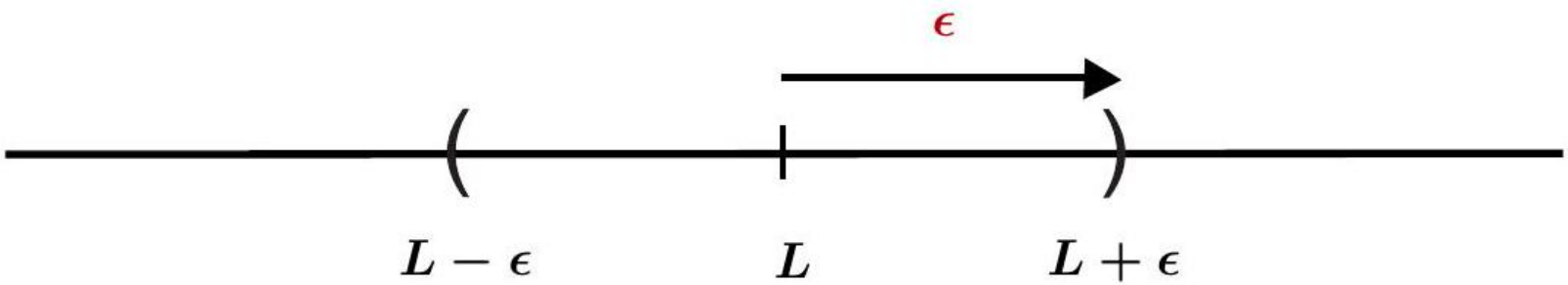


It is useful to look at how this works on the real line.

Assume $\lim_{n\to\infty} a_n = L$ and let $\epsilon > 0$.

We create an error band by moving ϵ units to the left from L to $L - \epsilon$,

The Limit of a Sequence

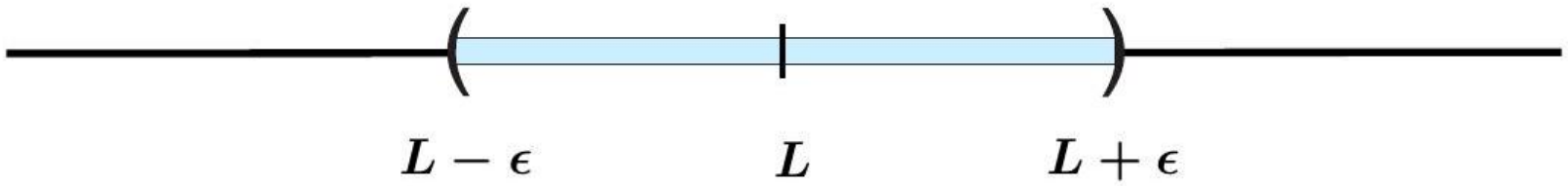


It is useful to look at how this works on the real line.

Assume $\lim_{n\to\infty} a_n = L$ and let $\epsilon > 0$.

We create an error band by moving ϵ units to the left from L to $L-\epsilon$, and then ϵ units to the right from L to $L+\epsilon$.

The Limit of a Sequence

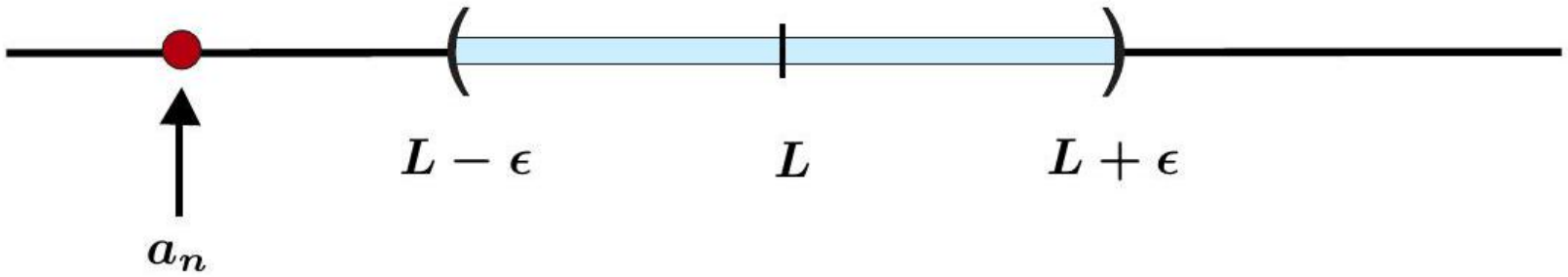


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We create an error band by moving ϵ units to the left from L to $L-\epsilon$, and then ϵ units to the right from L to $L+\epsilon$. This creates the interval $(L-\epsilon, L+\epsilon)$ as the "target."

The Limit of a Sequence



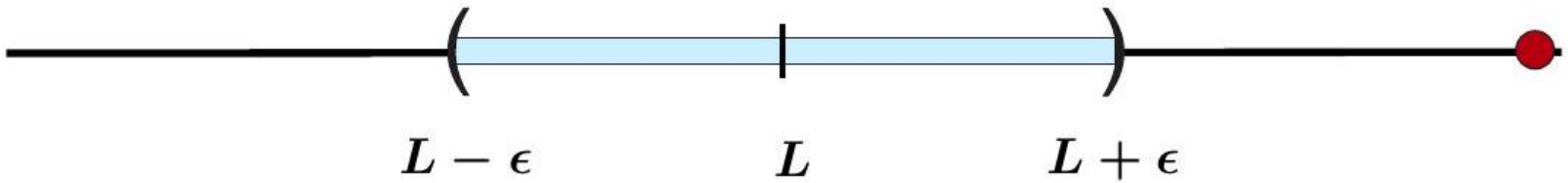
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Assume $\lim_{n\to\infty} a_n = L$ and let $\epsilon > 0$.

We create an error band by moving ϵ units to the left from L to $L-\epsilon$, and then ϵ units to the right from L to $L+\epsilon$. This creates the interval $(L-\epsilon, L+\epsilon)$ as the "target."

- Not all terms in $\{a_n\}$ must fall in $(L-\epsilon, L+\epsilon)$.

The Limit of a Sequence



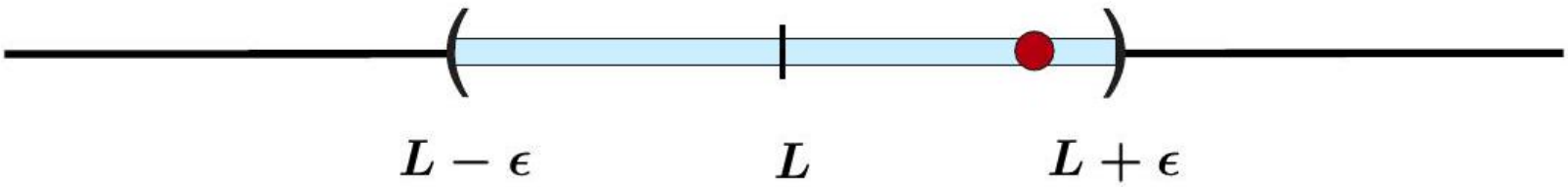
It is useful to look at how this works on the real line.

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We create an error band by moving ϵ units to the left from L to $L-\epsilon$, and then ϵ units to the right from L to $L+\epsilon$. This creates the interval $(L-\epsilon, L+\epsilon)$ as the "target."

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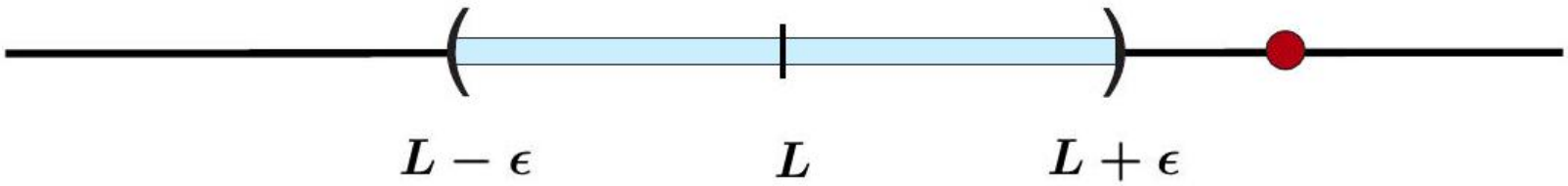
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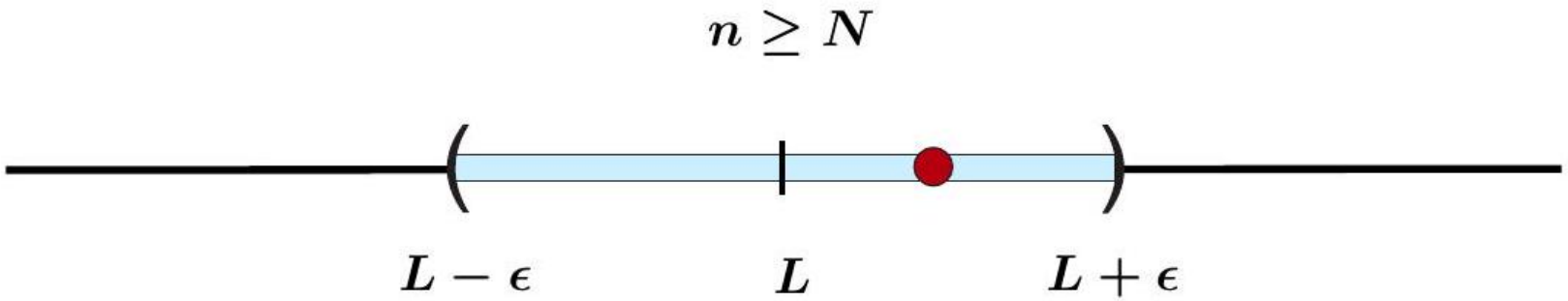
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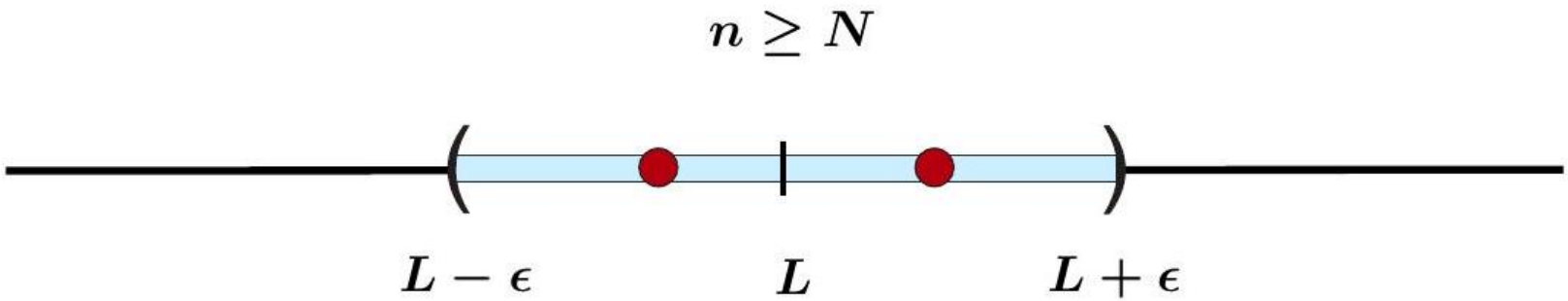
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- Not all terms in $\{a_n\}$ must fall in $(L - \epsilon, L + \epsilon)$.
- We can find $N \in \mathbb{N}$ such that if $n \geq N \Rightarrow a_n \in (L - \epsilon, L + \epsilon)$.

The Limit of a Sequence



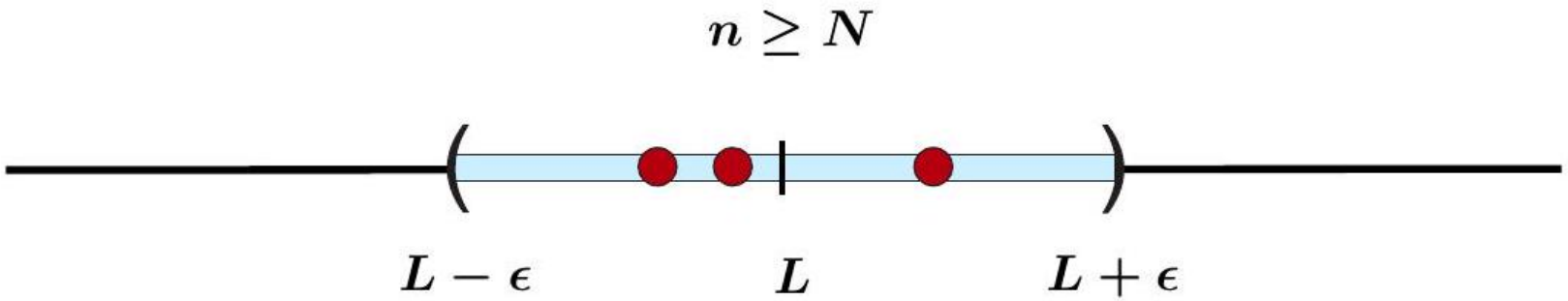
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The Limit of a Sequence



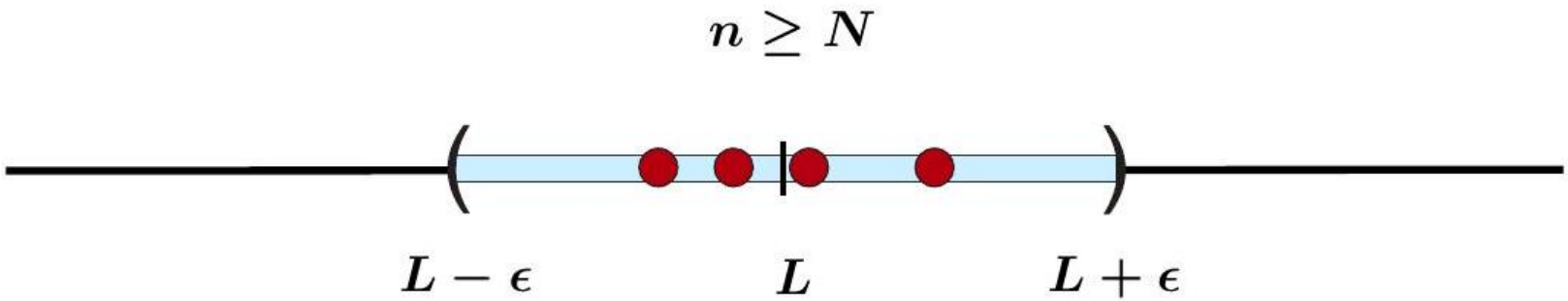
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The Limit of a Sequence



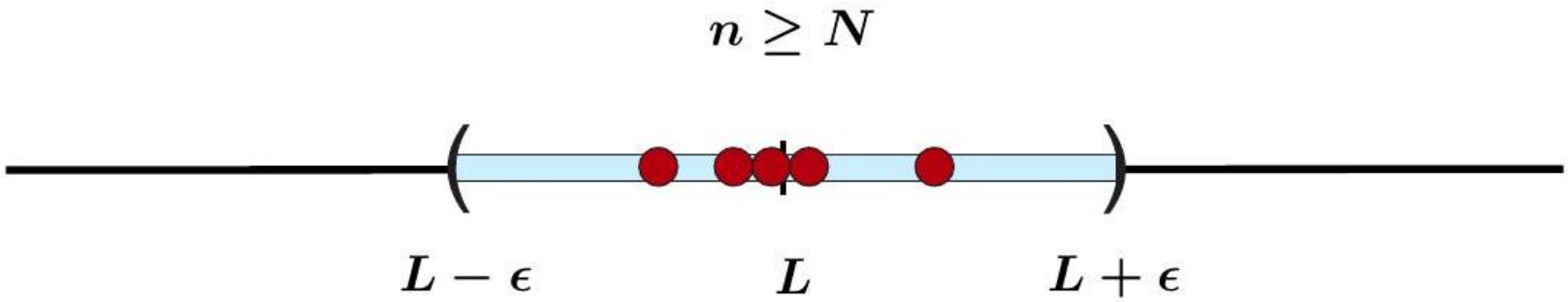
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- Not all terms in $\{a_n\}$ must fall in $(L - \epsilon, L + \epsilon)$.
- We can find $N \in \mathbb{N}$ such that if $n \geq N \Rightarrow a_n \in (L - \epsilon, L + \epsilon)$.

The Limit of a Sequence



It is useful to look at how this works on the real line.

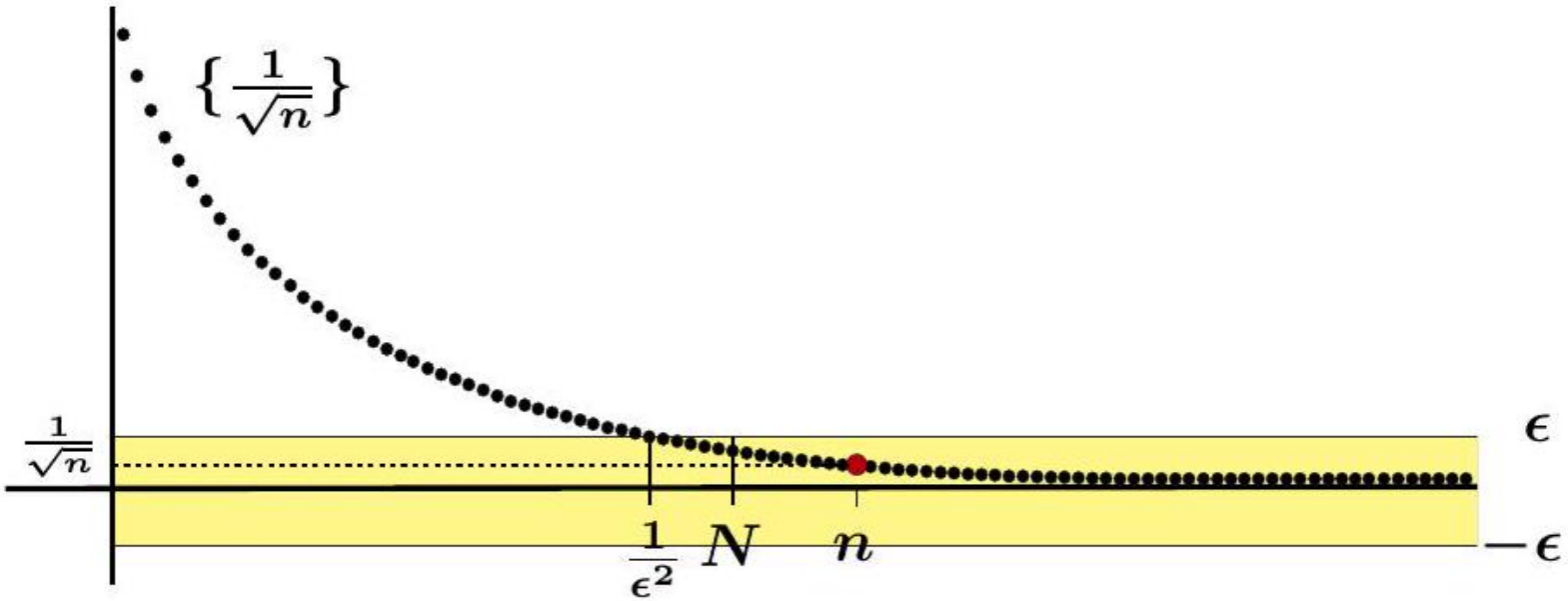
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- Not all terms in $\{a_n\}$ must fall in $(L - \epsilon, L + \epsilon)$.
- We can find $N \in \mathbb{N}$ such that if $n \geq N \Rightarrow a_n \in (L - \epsilon, L + \epsilon)$.

Example 1.2.3

Show that $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$.



Proof. Let $\epsilon > 0$. We need to find a cutoff N that satisfies the definition of the limit.

- If $\frac{1}{\epsilon^2} < n \Rightarrow \frac{1}{n} < \epsilon^2 \Rightarrow \frac{1}{\sqrt{n}} < \epsilon$.
- Hence, we can choose $N = \lceil \frac{1}{\epsilon^2} \rceil + 1$, then $n \geq N \Rightarrow \left| \frac{1}{\sqrt{n}} - 0 \right| \leq \frac{1}{\sqrt{N}} < \epsilon$.

Here $\lceil x \rceil$ denotes the smallest integer that is greater than or equal to x.

Example 1.2.4

Show that

$$\lim_{n\to\infty} \frac{3n+2}{4n+3} = \frac{3}{4}.$$

Solution: Given $\epsilon > 0$, we need to find a cutoff N so that if $n \geq N$, then

$$\left|\frac{3n+2}{4n+3} - \frac{3}{4}\right| < \epsilon.$$

Observe that

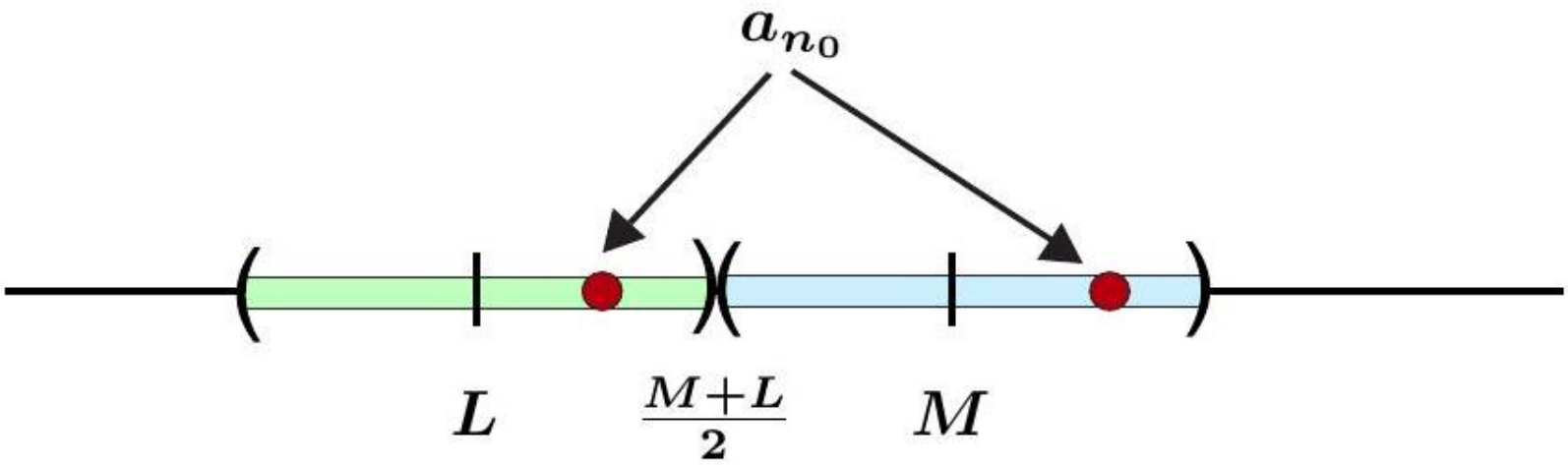
$$\left|\frac{3n+2}{4n+3} - \frac{3}{4}\right| = \left|\frac{12n+8}{16n+12} - \frac{12n+9}{16n+12}\right| = \left|\frac{-1}{16n+12}\right| = \frac{1}{16n+12}.$$

We want

$$\frac{1}{16n+12} < \epsilon \Rightarrow \frac{1}{\epsilon} < 16n+12 \Rightarrow \frac{1}{16\epsilon} - 0.75 < n,$$

so we can choose $N = \left\lceil \frac{1}{16\epsilon} - 0.75 \right\rceil$.

Uniqueness of Limits



Problem: Can $\{a_n\}$ have two different limits?

Assume $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} a_n = M$ with $L < M$.

Consider $\frac{M+L}{2}$. Let $\epsilon = \frac{M-L}{2}$.

Consider a_{n_0}. If n_0 is large enough, then

$$a_{n_0} \in (M-\epsilon, M+\epsilon)$$
$$a_{n_0} \in (L-\epsilon, L+\epsilon)$$

which is impossible!

Uniqueness of Limits

Theorem 1.2.5 (Uniqueness of Limits)

Assume that $\lim_{n\to\infty} a_n = L$ *and* $\lim_{n\to\infty} a_n = M$. *Then*

$$L = M.$$

Section 3

Arithmetic for Limits of Sequences

Problem:

Suppose that $a_n \to L$ and $b_n \to M$. What can we say about $c_n = 2a_n + b_n$?

We would hope that

$$\begin{aligned} c_n &= 2a_n + b_n \\ &\to 2L + M \end{aligned}$$

Arithmetic for Limits of Sequences

Theorem 1.3.1 (Arithmetic Rules for Limits of Sequences)

Let $\{a_n\}$ *and* $\{b_n\}$ *be sequences. Assume that* $\lim_{n\to\infty} a_n = L$ *and* $\lim_{n\to\infty} b_n = M$. *Then:*

i) For any $c \in \mathbb{R}, \lim_{n\to\infty} ca_n = c\boldsymbol{L}$.

ii) $\lim_{n\to\infty} (a_n + b_n) = L + M$.

iii) $\lim_{n\to\infty} a_n b_n = LM$.

iv) $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{L}{M}$ *if* $M \neq 0$.

Here we only prove (i) and (ii), and leave the proof of (iii) and (iv) as exercises.

Proof of (i):

For any $c \in \mathbb{R}$, if $\lim_{n\to\infty} a_n = L$, then $\lim_{n\to\infty} ca_n = cL$.

Case 1: $c = 0$

Let $c_n = c \cdot a_n$. Then

$$c_n = 0 \cdot a_n = 0$$

for all $\boldsymbol{n} \in \mathbb{N}$. So

$$c_n \to 0 = c \cdot L$$

Proof of (i) cont'd:

For any $c \in \mathbb{R}$, if $\lim_{n\to\infty} a_n = L$, then $\lim_{n\to\infty} ca_n = cL$.

Case 2: $c \neq 0$

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ so that if $n \geq N$, then

$$|a_n - L| < \frac{\epsilon}{|c|}$$

If $n \geq N$, then

$$\begin{aligned} |ca_n - cL| &= |c|\,|a_n - L| \\ &< |c|\frac{\epsilon}{|c|} \\ &= \epsilon. \end{aligned}$$

Remark: Our key observation in the above proof is

$$\text{Error of } [c \cdot \text{Quantity}] \approx c \cdot \text{Error}$$

Proof of (ii)

Assume that $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} b_n = M$. Then

$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n = L + M.$

Let $\epsilon > 0$. Choose $N_1 \in \mathbb{N}$ so that if $n \geq N_1$,

$$|a_n - L| < \frac{\epsilon}{2}.$$

Choose $N_2 \in \mathbb{N}$ so that if $n \geq N_2$,

$$|b_n - M| < \frac{\epsilon}{2}.$$

Let $N_0 = \max\{N_1, N_2\}$. If $n \geq N_0$, then

$$\begin{aligned} |(a_n + b_n) - (L + M)| &= |(a_n - L) + (b_n - M)| \\ &\leq |a_n - L| + |b_n - M| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon, \end{aligned}$$

concluding the proof of (ii).

Remark: Our key observation in the above proof is

$$\text{Error of a Sum} \approx \text{Sum of the Errors}$$

Limits of the Form $\frac{0}{0}$

Remark: Rule iv) says that if $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} b_n = M$, then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{L}{M}$$

if $M \neq 0$.

However, if $\lim_{n\to\infty} b_n = 0$, then $\lim_{n\to\infty} \frac{a_n}{b_n}$ may or may not exist.

Example 1.3.2

- $a_n = \frac{1}{n}, b_n = \frac{1}{n} \Rightarrow \frac{a_n}{b_n} = 1 \to 1.$
- $a_n = \frac{1}{n}, b_n = \frac{1}{n^2} \Rightarrow \frac{a_n}{b_n} = \frac{\frac{1}{n}}{\frac{1}{n^2}} = n \to \infty.$
- $a_n = \frac{1}{n^2}, b_n = \frac{1}{n} \Rightarrow \frac{a_n}{b_n} = \frac{\frac{1}{n^2}}{\frac{1}{n}} = \frac{1}{n} \to 0.$

Example 1.3.3

Evaluate $\lim_{n\to\infty} \frac{3n^2+2n-1}{4n^2+2}$.

Observation: Note that both $\{3n^2+2n-1\}$ and $\{4n^2+2\}$ diverge.

For large n,

$$3n^2+2n-1 \approx 3n^2 \quad \text{and} \quad 4n^2+2 \approx 4n^2,$$

so

$$\frac{3n^2+2n-1}{4n^2+2} \approx \frac{3n^2}{4n^2} = \frac{3}{4}.$$

This might lead us to guess:

$$\lim_{n\to\infty} \frac{3n^2+2n-1}{4n^2+2} = \frac{3}{4}.$$

Example: rigorous derivation

Example: Evaluate $\lim_{n\to\infty} \frac{3n^2+2n-1}{4n^2+2}$.

Solution.

$$\begin{aligned}
\lim_{n\to\infty} \frac{3n^2+2n-1}{4n^2+2} &= \lim_{n\to\infty} \frac{n^2}{n^2} \cdot \frac{3+\frac{2}{n}-\frac{1}{n^2}}{4+\frac{2}{n^2}} \\
&= \lim_{n\to\infty} \frac{3+\frac{2}{n}-\frac{1}{n^2}}{4+\frac{2}{n^2}} \\
&= \frac{\lim_{n\to\infty} 3 + \lim_{n\to\infty} \frac{2}{n} - \lim_{n\to\infty} \frac{1}{n^2}}{\lim_{n\to\infty} 4 + \lim_{n\to\infty} \frac{2}{n^2}} \\
&= \frac{3+2(0)-0}{4+2(0)} \\
&= \frac{3}{4}.
\end{aligned}$$

Example 1.3.4

Evaluate $\lim_{n\to\infty} \frac{3n^2+2n-1}{4n^3+2}$.

Observation: For large values of n the terms in the sequence behave like

$$\frac{3n^2}{4n^3} = \frac{3}{4n}$$

which converges to 0.

Example: rigorous derivation

Example: Evaluate $\lim_{n\to\infty} \frac{3n^2+2n-1}{4n^3+2}$.

Solution.

$$\begin{aligned}
\lim_{n\to\infty} \frac{3n^2+2n-1}{4n^3+2} &= \lim_{n\to\infty} \frac{n^2}{n^3} \cdot \frac{3+\frac{2}{n}-\frac{1}{n^2}}{4+\frac{2}{n^3}} \\
&= \lim_{n\to\infty} \frac{1}{n} \cdot \frac{3+\frac{2}{n}-\frac{1}{n^2}}{4+\frac{2}{n^3}} \\
&= \lim_{n\to\infty} \frac{1}{n} \cdot \lim_{n\to\infty} \frac{3+\frac{2}{n}-\frac{1}{n^2}}{4+\frac{2}{n^3}} \\
&= 0 \cdot \frac{3}{4} \\
&= 0.
\end{aligned}$$

Example 1.3.5

Show that $\left\{\frac{3n^2+5}{n^{3/2}+2}\right\}$ diverges.

Observation: For large values of n the terms in the sequence behave like

$$\frac{3n^2}{n^{3/2}} = 3n^{\frac{1}{2}}$$

which diverges.

We have

$$\begin{aligned}\lim_{n\to\infty}\frac{3n^2+5}{n^{3/2}+2} &= \lim_{n\to\infty}\frac{n^2}{n^{\frac{3}{2}}}\cdot\frac{3+\frac{5}{n^2}}{1+\frac{2}{n^{3/2}}}\\ &= \lim_{n\to\infty}\sqrt{n}\cdot\frac{3+\frac{5}{n^2}}{1+\frac{2}{n^{3/2}}}\\ &= \infty.\end{aligned}$$

Limits of the Form $\frac{p(n)}{q(n)}$

Let

$$a_n = \frac{b_0 + b_1 n + b_2 n^2 + b_3 n^3 + \cdots + b_j n^j}{c_0 + c_1 n + c_2 n^2 + \cdots + c_k n^k}$$

Suppose $b_j \neq 0$ and $c_k \neq 0$.

By factoring out n^j from the numerator and n^k from the denominator, and then rewriting the sequence as

$$a_n = \frac{n^j}{n^k}\left[\frac{\frac{b_0}{n^j} + \frac{b_1}{n^{j-1}} + \frac{b_2}{n^{j-2}} + \frac{b_3}{n^{j-3}} + \cdots + b_j}{\frac{c_0}{n^k} + \frac{c_1}{n^{k-1}} + \frac{c_2}{n^{k-2}} + \cdots + c_k}\right]$$

we can conclude that

$$\lim_{n\to\infty} a_n = \begin{cases} \dfrac{b_j}{c_k} & \text{if } j = k \\ 0 & \text{if } j < k \\ \text{does not exist} & \text{if } j > k \end{cases}$$

Example 1.3.6

Evaluate

$$\lim_{n\to\infty} \sqrt{n^2+n} - n$$

Note:

i) $\lim_{n\to\infty} \sqrt{n^2+n} - \lim_{n\to\infty} n = \infty - \infty$ has no meaning.

ii) $a_{100} = 0.498756211$, $a_{10^6} = 0.499999875$
$\Rightarrow \lim_{n\to\infty} \sqrt{n^2+n} - n = \frac{1}{2}$?

We have

$$\begin{aligned}
\sqrt{n^2+n} - n &= \left(\sqrt{n^2+n} - n\right) \cdot \frac{\sqrt{n^2+n}+n}{\sqrt{n^2+n}+n} \\
&= \frac{\left(n^2+n\right) - n^2}{\sqrt{n^2+n}+n} = \frac{n}{\sqrt{n^2+n}+n} \\
&= \frac{n}{n} \cdot \frac{1}{\sqrt{1+\frac{1}{n}}+1} = \frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2}.
\end{aligned}$$

Squeeze Theorem for Sequences

Example: Consider the sequence $\left\{\frac{\sin(n)}{n}\right\}$.

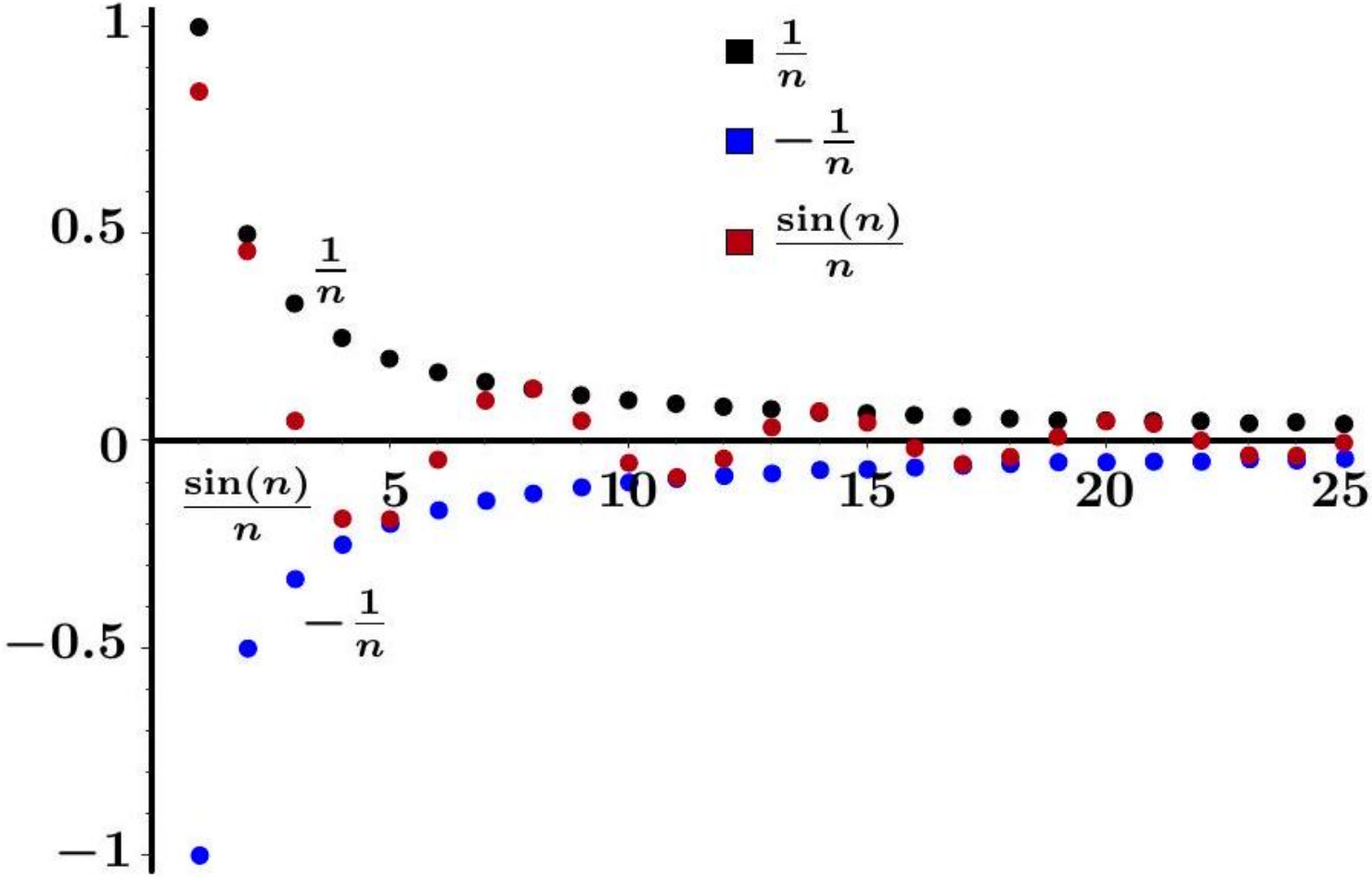
Does it converge? If so, what does it converge to?

Note that $|\sin(n)| \leq 1$ for all n, so

$$\left|\frac{\sin(n)}{n}\right| \leq \frac{1}{n}$$

Then

$$-\frac{1}{n} \leq \frac{\sin(n)}{n} \leq \frac{1}{n}$$



$$\lim_{n\to\infty} \frac{1}{n} = 0 = \lim_{n\to\infty} -\frac{1}{n} \Rightarrow \lim_{n\to\infty} \frac{\sin(n)}{n} = 0.$$

Squeeze Theorem for Sequences

Theorem 1.3.7 (Squeeze Theorem for Sequences)

Assume that $\{a_n\}, \{b_n\}$, *and* $\{c_n\}$ *are such that*

$$c_n \leq b_n \leq a_n$$

for all $n \in \mathbb{N}$.
Assume also that

$$\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n.$$

Then $\lim_{n\to\infty} b_n$ *exists and*

$$\lim_{n\to\infty} b_n = L.$$

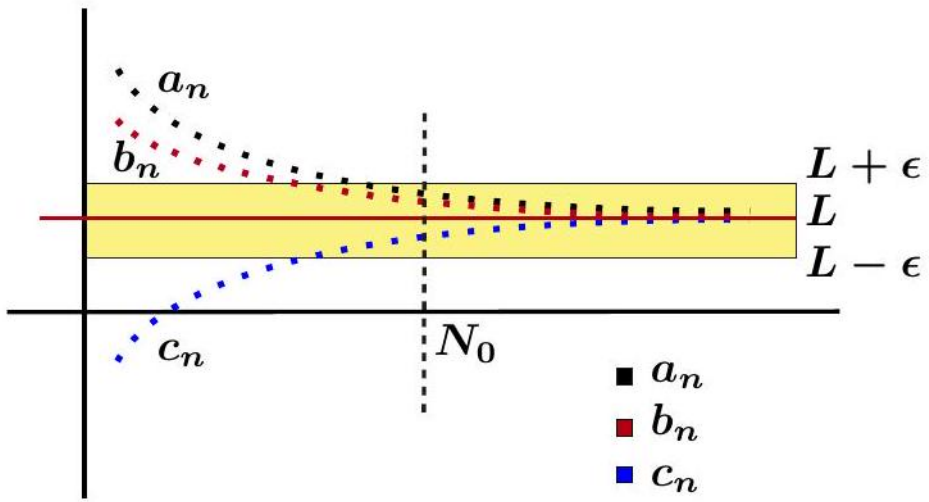
Proof of Squeeze Theorem for Sequences

Consider the sequences
$\{a_n\}$ and $\{c_n\}$. Assume that

$$\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n$$

Let $\epsilon > 0$.

We can
find an N_0 such that if $n \geq N_0$, then



$$L - \epsilon < c_n \leq a_n < L + \epsilon$$

Therefore, if $n \geq N_0$, then

$$L - \epsilon < c_n \leq b_n \leq a_n < L + \epsilon \Rightarrow |b_n - L| < \epsilon.$$

Section 4

Monotone Convergence Theorem

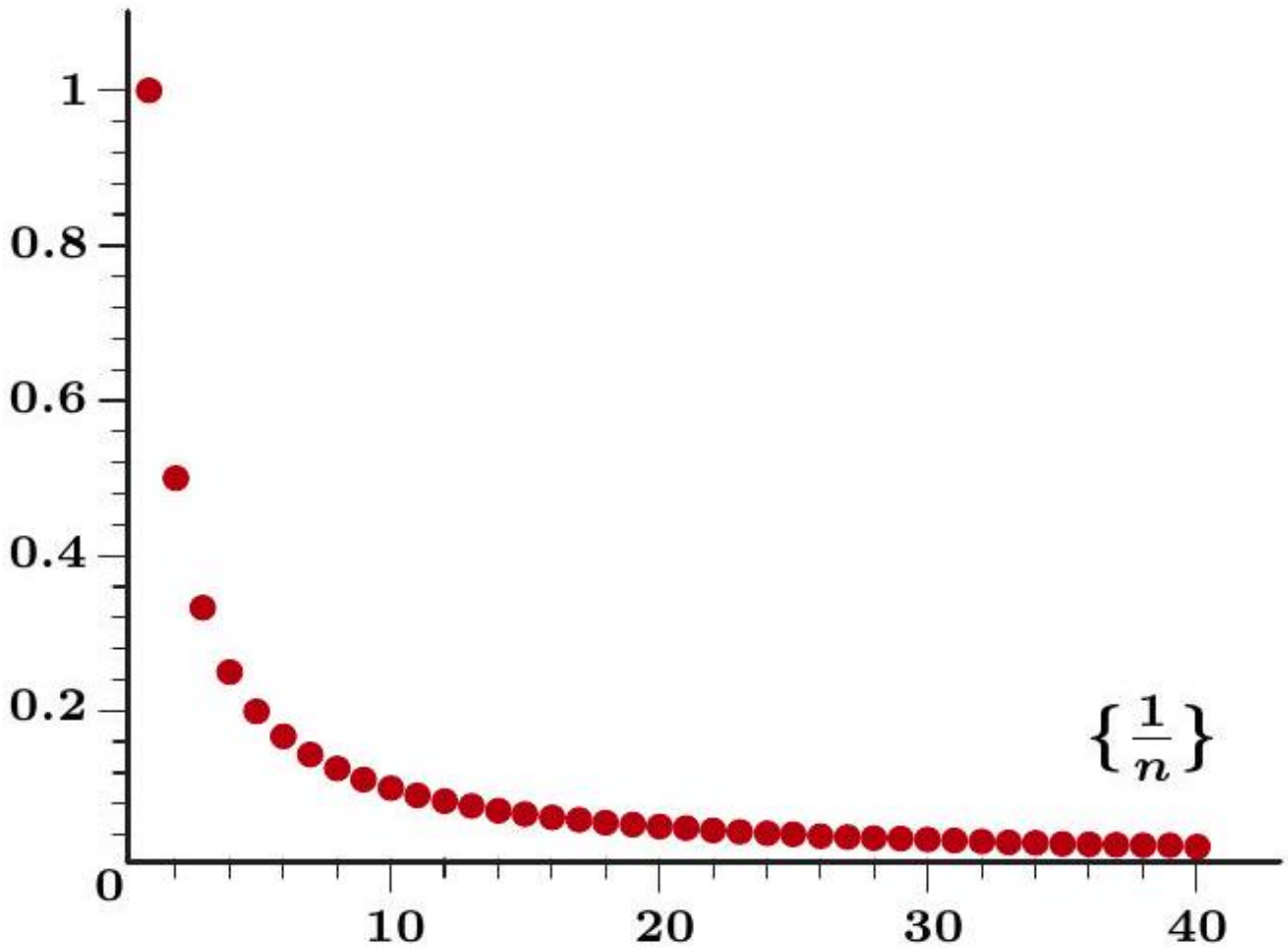
Monotonic Sequences

Definition 1.4.1 (Monotonic Sequences)

We say that a sequence $\{a_n\}$ is:

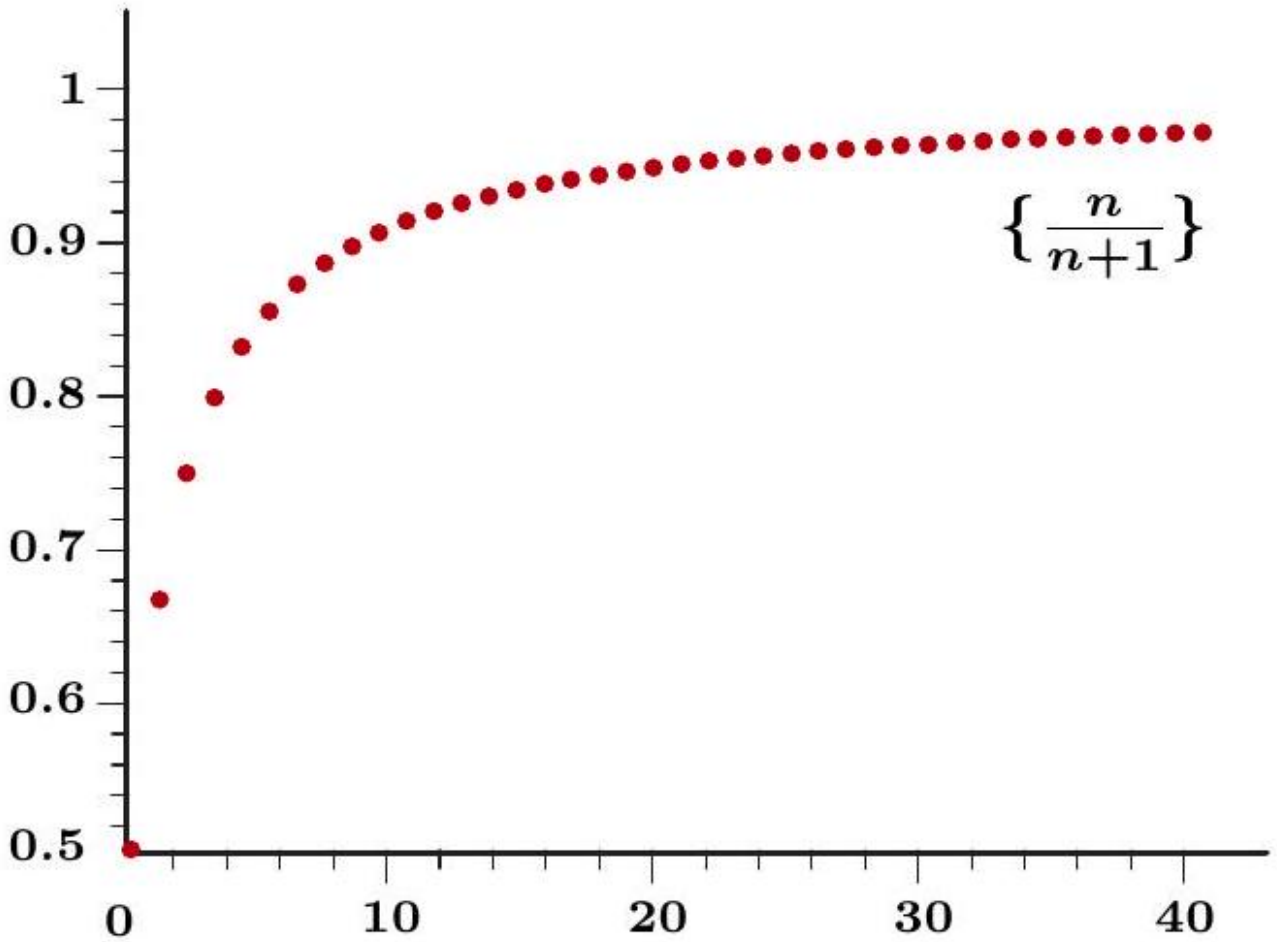
- strictly increasing if $a_n < a_{n+1}$, for all $n \in \mathbb{N}$.
- increasing if $a_n \leq a_{n+1}$, for all $n \in \mathbb{N}$.
- strictly decreasing if $a_n > a_{n+1}$, for all $n \in \mathbb{N}$.
- decreasing if $a_n \geq a_{n+1}$, for all $n \in \mathbb{N}$.
- monotonic if $\{a_n\}$ is either decreasing or increasing.

Examples of Monotonic Sequences



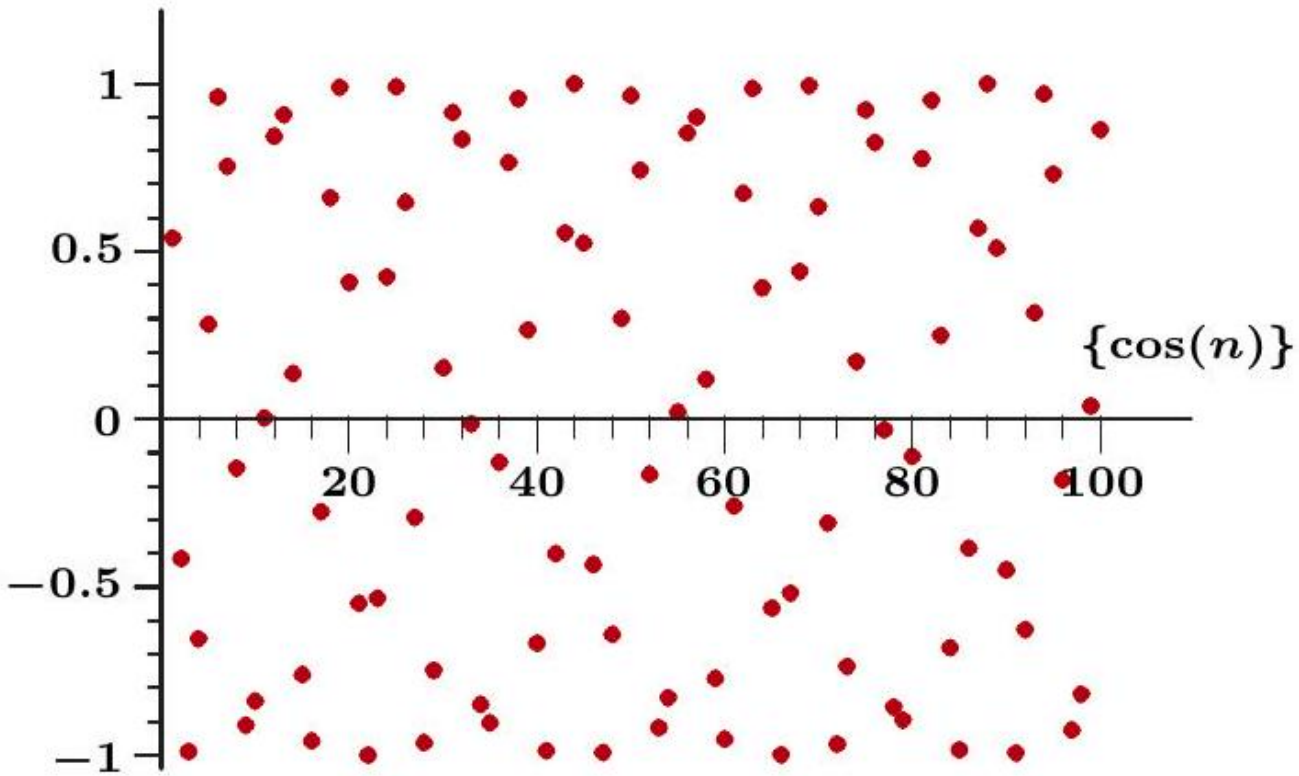
- The sequence $\left\{\frac{1}{n}\right\}$ is decreasing.

Examples of Monotonic Sequences



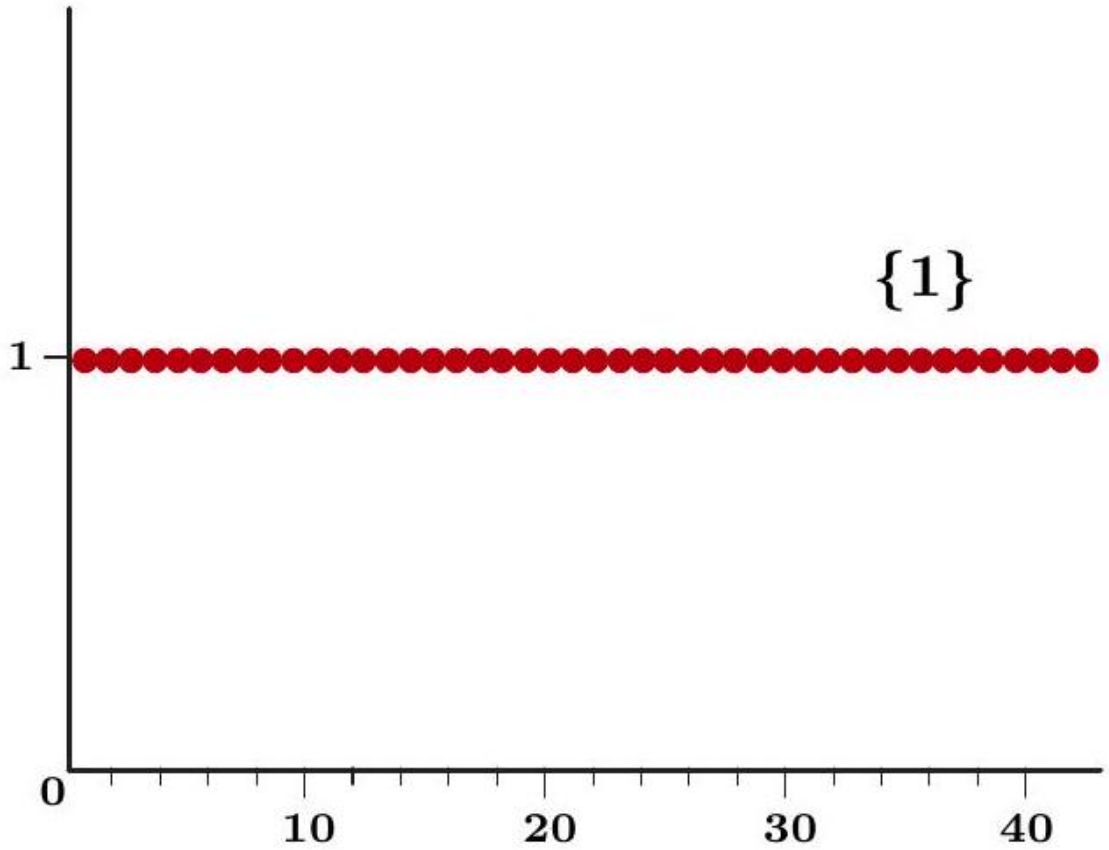
- The sequence $\left\{\frac{n}{n+1}\right\} = \left\{1 - \frac{1}{n+1}\right\}$ is increasing.

Examples of Monotonic Sequences



- The sequence $\{\cos(n)\}$ is neither decreasing or increasing.

Examples of Monotonic Sequences



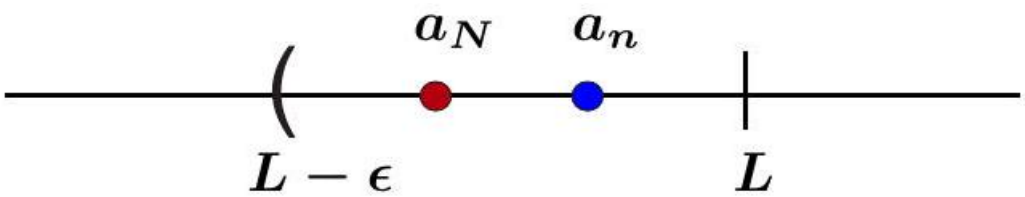
- The constant sequence $\{1\}$ is both decreasing and increasing.

Monotone Convergence Theorem (MCT)

Theorem 1.4.2 (Monotone Convergence)

If $\{a_n\}$ is increasing and bounded above, then $\{a_n\}$ converges to $L = lub\,\{a_n\}$.

Proof of MCT



Assume that $\{a_n\}$ is increasing and bounded above with $L = lub\{a_n\}$. Let $\epsilon > 0$. Then $L - \epsilon < L$, so $L - \epsilon$ is not an upper bound of $\{a_n\}$. Hence, we can find an $N \in \mathbb{N}$ such that $L - \epsilon < a_N$.

If $n \geq N$, then

$$L - \epsilon < a_N \leq a_n \leq L.$$

This shows that if $n \geq N$, then $|a_n - L| < \epsilon$. So

$$\lim_{n\to\infty} a_n = L.$$

Monotone Convergence Theorem

Note: We can also show that:

- if $\{a_n\}$ is decreasing and bounded below, then

$$\lim_{n\to\infty} a_n = glb\,\{a_n\}$$

Example 1.4.3

We are ready to show that

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$$

exists.

In fact, let $a_n = \left(1+\frac{1}{n}\right)^n$. Use the binomial expansion,

$$\begin{aligned}
a_n =& 1 + n\frac{1}{n} + \frac{n(n-1)}{2!}\frac{1}{n^2} + \ldots + \frac{n(n-1)\ldots 3\cdot 2\cdot 1}{n!}\frac{1}{n^n} \\
=& 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n}\right) + \ldots + \frac{1}{n!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{n-1}{n}\right) \\
a_{n+1} =& 1 + 1 + \frac{1}{2!}\left(1-\frac{1}{n+1}\right) + \ldots + \frac{1}{n!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right)\ldots\left(1-\frac{n-1}{n+1}\right) \\
& + \frac{1}{(n+1)!}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right)\ldots\left(1-\frac{n}{n+1}\right).
\end{aligned}$$

Since $1-\frac{i}{n} < 1-\frac{i}{n+1}$, we have $a_n < a_{n+1}$. Thus $\{a_n\}$ is increasing.

Example cont'd

To show $\{a_n\}$ is bounded, we see that

$$\begin{aligned} a_n &< 1 + 1 + \frac{1}{2!} + \ldots + \frac{1}{n!} \\ &< 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \ldots + \frac{1}{(n-1)n} \\ &= 1 + 1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \ldots + \left(\frac{1}{n-1} - \frac{1}{n}\right) \\ &= 1 + 1 + 1 - \frac{1}{n} < 3. \end{aligned}$$

Thus the sequence $\{a_n\}$ is bounded:

$$a_1 = 2 \leq a_n < 3, \quad n \geq 1.$$

By MCT, the sequence $\{a_n\}$ converges.

The number e

Denote the limit to be e:

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e.$$

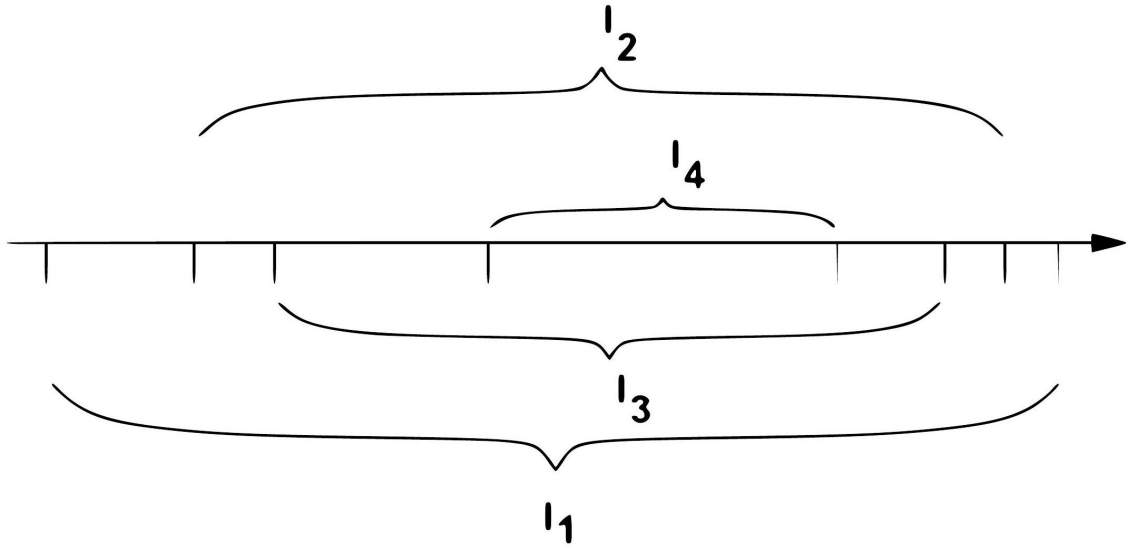
The Squeeze Theorem implies that $2 \leq e \leq 3$. In fact,

$$e = 2.71828\ldots$$

Section 5

Nested Intervals*

We can use the monotone convergence theorem to prove the nested interval theorem.



Theorem 1.5.1 (Nested interval theorem)

If $I_1, I_2, I_3, \ldots$ is a sequence of closed intervals that are "nested," that is, each I_n contains I_{n+1}, then the intervals I_n have at least one point in common. If the lengths of the intervals tend to 0, there is exactly one point in common.

Proof

Denote by a_n and b_n the left and right endpoints of I_n.

The assumption of nesting means that the sequence $\{a_n\}$ increases, the sequence $\{b_n\}$ decreases, and that each b_n is greater than every a_n. So by the monotone convergence theorem, $\{a_n\}$ converges to some limit a, and $\{b_n\}$ to some limit b.

By the way we constructed the limits, we know that $a_n \leq a$ and $b \leq b_n$.

Proof cont'd

Now, a cannot be greater than b, for otherwise, some a_n would be larger than some b_m, violating the nesting assumption. So a must be less than or equal to b, and $a_n \leq a \leq b \leq b_n$.

If the lengths of the intervals I_n tend to zero as n tends to infinity, then the distance between a and b must be zero, and the intervals I_n have exactly one point in common, namely the point $a = b$.

If the lengths of the I_n do not tend to zero, then $[a, b]$ belongs to all the intervals I_n, and the intervals have many points in common.

Section 6

Cauchy Sequences*

Cauchy Sequences

Definition 1.6.1

A sequence of numbers $\{a_n\}$ is called a Cauchy sequence if given any tolerance ϵ, no matter how small, there is an integer N such that for all integers n and m greater than N, a_n and a_m differ from each other by less than ϵ.

Example 1.6.2

Verify directly that the sequence $\left\{\frac{1}{n}\right\}$ is a Cauchy sequence.

Let ϵ be any tolerance and let N be a positive integer greater than $\frac{2}{\epsilon}$. If $m > N$ and $n > N$, then by the triangle inequality,

$$\left|\frac{1}{m} - \frac{1}{n}\right| \leq \frac{1}{m} + \frac{1}{n} \leq \frac{1}{N} + \frac{1}{N} < \epsilon.$$

So $\left\{\frac{1}{n}\right\}$ is a Cauchy sequence.

Theorem 1.6.3 (The Cauchy criterion)

A sequence is convergent if and only if it's a Cauchy sequence.

To show that a convergent sequence is Cauchy is easy. To show that a Cauchy sequence converges, we proceed in four steps:

1. First, we show that every sequence has a monotone subsequence. This first step is worth recognizing as a "lemma", a key stepping-stone in the argument.
2. Second, we show that every Cauchy sequence is bounded (and hence every subsequence as well).
3. Third, we recognize that a Cauchy sequence has a monotone subsequence that is bounded and therefore converges.
4. Fourth, we show that a Cauchy sequence converges to the same limit as its monotone subsequence.

We mentioned subsequences on the previous slide, whose exact definition is as follows.

Definition 1.6.4 (Subsequence)

Given a sequence $\{a_n\}$, picking out infinitely many terms of $\{a_n\}$, we get a new sequence. To be precise, given

$$n_1 < n_2 < \ldots < n_k < \ldots$$

where each n_k is a positive integer, we can form a new sequence $\{b_k\}$, where $b_k = a_{n_k}$ for each k, called a subsequence of $\{a_n\}$.

Example 1.6.5

Consider the sequence $\{a_n = (-1)^n\}$.

- We see that $\{a_{2k}\}$ is a subsequence, each term of which is the constant 1. Thus, $\lim_{k\to\infty} a_{2k} = 1$.
- Another subsequence $\{a_{2k+1}\}$, each term of which is -1, has a limit: $\lim_{k\to\infty} a_{2k+1} = -1$.

We see that the limits of the two subsequences are different. We also knew in advance that the sequence $\{a_n\}$ is divergent.

Lemma 1.6.6

Every infinite sequence of numbers has an infinite monotonic subsequence.

Proof (of the Lemma). Let $a_1, a_2, \ldots$ be any sequence. We shall show that it contains either an increasing or a decreasing subsequence.

We start by trying to construct an increasing subsequence. Start with a_1 and take any subsequent term in the original sequence that is greater than or equal to a_1 as the next term of the subsequence. Continue in this fashion. If we can continue indefinitely, we have the desired increasing subsequence.

Proof of the Lemma cont'd

Suppose, on the other hand, that we get stuck after a finite number of steps at a_j because $a_n < a_j$ for all $n > j$. Then we try again to construct an increasing sequence, starting this time with a_{j+1}. If we can continue indefinitely, we have an increasing subsequence. If, on the other hand, we get stuck at a_k because $a_n < a_k$ for all $n > k$, then we can again try to construct an increasing subsequence starting at a_{k+1}.

Proceeding in this fashion, we either have success at some point, or an infinite sequence of failures. In the second case, the sequence of points $a_j, a_k, \ldots$ where the failures occurred constitutes a decreasing subsequence. This completes the proof of the lemma.

Proof of the Theorem

Assume $\{a_n\}$ is convergent to L. For any $\epsilon > 0$, there exists an integer $N > 0$ such that whenever $n \geq N, |a_n - L| < \epsilon/2$. Therefore, when $n, m \geq N$,

$$|a_n - a_m| = |(a_n - L) - (a_m - L)| \leq |a_n - L| + |a_m - L| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Conversely, assume $\{a_n\}$ is a Cauchy sequence. The lemma we just proved guarantees that a Cauchy sequence has a monotone subsequence.

Next, we show that a Cauchy sequence is bounded. Being Cauchy ensures that there exists an N such that the terms from a_N onward are all within 1 of each other. This means that the largest of the numbers

$$1 + a_1, 1 + a_2, \ldots, 1 + a_N$$

is an upper bound for $\{a_n\}$, and the smallest of the numbers

$$-1 + a_1, -1 + a_2, \ldots, -1 + a_N$$

is a lower bound of $\{a_n\}$.

Proof of the Theorem cont'd

Now by the monotone convergence theorem, the monotone subsequence of $\{a_n\}$ converges to a limit a.

Next we show that not only a subsequence but the whole sequence converges to a. Let us write a_m for an element of the subsequence and a_n for any element of the sequence, and $a - a_n = a - a_m + a_m - a_n$. The triangle inequality gives

$$|a - a_n| \leq |a - a_m| + |a_m - a_n| .$$

The first term on the right is less than any prescribed ϵ for m large, because the subsequence converges to a. The second is less than any prescribed ϵ for m and n both large, because this is a Cauchy sequence. This proves that the whole sequence also converges to a. This completes the proof of the theorem.

Example 1.6.7

Show that the sequence $\{x_n\}$, where

$$x_n = 1 + \frac{1}{2} + \ldots + \frac{1}{n}$$

is divergent.

We use the Cauchy Criterion. In fact, for any n,

$$|x_{2n} - x_n| = \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{n+n} \geq \frac{1}{n+n} + \frac{1}{n+n} + \ldots + \frac{1}{n+n} = \frac{1}{2}$$

Thus, if we choose $\epsilon = \frac{1}{2}$, then for any N, the inequality

$$|x_n - x_m| \geq \epsilon$$

holds for all $m = 2n$, even $n, m \geq N$. Therefore, by the Cauchy Criterion, the sequence $\{x_n\}$ diverges.

Example 1.6.8

Show that the sequence $\{x_n\}$, where

$$x_n = 1 + \frac{1}{2^2} + \ldots + \frac{1}{n^2}$$

is convergent.

We use the Cauchy Criterion. In fact, if we write $m = n + k$ where $k > 0$, then

$$\begin{aligned}
&|x_m - x_n| \\
=&\frac{1}{(n+1)^n} + \frac{1}{(n+2)^2} + \ldots + \frac{1}{(n+k)^2} \\
\leq&\frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)} + \ldots + \frac{1}{(n+k-1)(n+k)} \\
=&\left(\frac{1}{n} - \frac{1}{n+1}\right) + \left(\frac{1}{n+1} - \frac{1}{n+2}\right) + \ldots + \left(\frac{1}{n+k-1} - \frac{1}{n+k}\right) \\
=&\frac{1}{n} - \frac{1}{n+k}.
\end{aligned}$$

Example cont'd

Thus, for any ϵ, choose N to be any positive integer greater than $1/\epsilon$. Whenever $n \geq N$ and $m \geq N$, we have

$$|x_n - x_m| \leq \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

By the Cauchy Criterion, the sequence $\{x_n\}$ converges.